# DOUBLE JEOPARDY: The Vulnerability Spiral, Quantifying Compound Coastal Risk and Governance Evidence Alignment Across 5 Islands

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## Entry 1

### Project Overview

I Am running the DOUBLE JEOPARDY project to check a big danger. (See, the main issue is tht rising sea levels hit small islands badly because their natural walls like trees and rocks are dying). I did not want to put all coastal green stuff into jst one big group. So, I chose to track mangroves nd coral reefs completely separately. My data comes from 5 different islands across 3 big ocean basins: Maldives, Lakshadweep, Seychelles, Fiji, nd the Canary Islands. Calculated the health paths for corals n mangroves as 2 different things instead of mixing them into one single score. Finally, I mixed the coral number with the real physical danger data to make one big final score, keeping mangrove reported separately since it never showed a decline to begin with.

I am also looking at governance. (Basically, I want to see if officially making a place a protected area, which is wht big global groups nd island leaders always say they are doing to control the danger to the beach, actually matches up with the spots where my numbers show the highest real danger). Chose to check this because it moves my work away from jst basic science. Am testing if government rules actually follow the real field proof or if they are jst totally disconnected from each other. My findings will matter a lot for things like adaptation finance, environmental economics, and marine spatial governance.

## Problem Statement

Big money groups and climate policy makers think all beach ecosystems are jst one single safety category. They assume tht mangroves and coral reefs will always die together at the same time n jointly decide how badly a village gets hit by sea level rise, but nobody really tests this old idea with proper independent data over long timelines on a proper big cross national scale. So I think this is a huge mistake..... (If these 2 items are actually dying at different speeds or going in totally opposite directions, then putting them in one single bucket will make leaders spend money in the wrong places nd completely miscalculate which coastal zones are in real trouble).

Then there is a 2nd major issue on top of tht. Global cash frameworks n national laws use the "protected area" tag as a fake proof tht the ecosystem is being managed nicely..... (In reality, people draw these protection borders because of old politics or administrative ease instead of checking new, live risk numbers). I found tht nobody has used independent satellite evidence before my project to check if these 5 islands hve real safety or just "paper parks" tht are totally disconnected from the real danger zones. So I wanted to see if the protection lines actually match the highest ecological nd physical vulnerability.....

## Aim

I am building a reproducible geospatial framework. (My main goal here is to separately count the actual physical danger from sea level rise nd track how mangroves nd coral reefs are changing over multiple time periods across 5 different islands so I can finally create a real, evidence based composite vulnerability assessment). Then I will do a 2nd step. I will run stats to check if the current protected area governance lines actually match up with the real danger spots I found, or if there is a massive governance evidence gap happening on the ground.

## Research Questions

My 1st question is about nature..... (Basically, I am trying to find out if the real physical danger of sea level rise mixes up with dying natural borders to create a much worse total danger for villages across 5 islands, nd if this whole process looks exactly identical for every ecosystem type or if it actually depends on whether mangroves or corals are the ones rotting away in that specific spot). So I need to solve this doubt 1st.

Then I have my 2nd question. Want to check if the actual places where governments put official WDPA protected area tags match my field data on high danger, or if these official marks are jst completely independent of (or maybe even opposite to) the real ecosystem risks I found using satellite maps.

## Hypotheses

My 1st guess is abt coral reefs. (Think my long term stats will show a clear increase in thermal bleaching stress over many decades, especially on islands that face much higher physical sea level rise exposure). Will check this 1st.

Next, I look at mangroves. I think my data will show they are shrinking in total area over this exact same time frame as a totally separate problem alongside the coral dying. (I'll test this the same way as the 1st one nd log whatever the charts show).

Finally, I hve a governance guess. Expect tht islands n coastlines with high total vulnerability scores will have more protected area coverage because rulers should react to real danger. But if I find zero connection or the

complete opposite thing, tht is still a real result proving tht protection rules in my sample are not based on actual risk data, n I will report it straight either way.....

## Objectives

1st, I will build a physical exposure dataset for villages across 5 islands. (Am using basic height nd slope map data to calculate exactly how many houses n local settlements sit below the dangerous one meter sea level rise mark). This is my starting point.

Then I will download the full Global Mangrove Watch archive. I chose to track real area changes across several separate years instead of jst doing a lazy before n after comparison. Next, I need to check the coral reefs. (I am using satellite Degree Heating Week data to map out if bleaching heat stress is actually rising over time).

After tht, I will mix everything. I am combining my physical danger maps and ecosystem loss data to calculate a single Compound Vulnerability Score for every island..... Then I am going to check the official WDPA protected area maps. Will run a stats test to see if these government lines actually match my final vulnerability scores.....

I also want to track new buildings..... (I will use satellite maps to do built up area change detection to see if houses and infrastructure are expanding right inside or nxt to the fragile ecosystem buffer zones). Am going to double check every single trend using multiple independent time points so my data stays clean. Finally, I will package all these scripts into a reproducible geospatial framework n an interactive dashboard to show my risk findings n governance analysis clearly to adaptation policy and conservation finance people.

## Methodology Summary

Figured out physical exposure 1st. (Basically, I took the height n slope map data and crossed it with where people actually live on all 5 islands so I could calculate the percentage of houses sitting right at or below the one meter line). This gave me the main numbers.

Then I looked at the nature data. For mangroves, I chose to use multiple independent time points from the Global Mangrove Watch archive because I wanted to check for real area changes over a few decades. Then I checked the corals..... (Took a continuous multi decadal satellite thermal stress time series n simply compared the old averages against the recent ones to see how bad things got).....

Finally, I mixed the physical exposure numbers with the coral signal to calculate a single, normalized Compound Vulnerability Score for each island. Mangrove stayed out of tht score completely, since it showed no decline nd I didnt want it just sitting there as a flat zero inside a weighted formula — it stays reported as its own separate standalone result instead.

Next, I am coding the governance part..... (So I counted the total size nd density of official protected areas using data from the World Database on Protected Areas for every island n even for single coastal zones where the map resolution was good enough). Ran a stats test after that..... I directly compared those protection numbers against my final Compound Vulnerability Score because I want to see if high risk zones are more likely, equally likely, or less likely to get official government protection. This step is very important to me. (Instead of just writing generic comments abt politics at the very end of my paper, I chose to convert my nature findings into a real, testable statement to check if government policies actually work on the ground).

So I will do a later phase. (For this part, I plan to use satellite based built up area change detection to map out if new houses, roads, nd village expansions are heavily building up right inside or close to the fragile

ecosystem buffer zones). So I want to test a weird loop..... Think humans reacting to environment pressures might actually end up making the exact same vulnerability worse by destroying their own natural shields. Then I have one more check. (So I am going to calculate if this bad building encroachment happens more often inside official protected boundaries or completely outside of them).....

### Study Area

I picked 5 islands. (My list spreads across 3 ocean basins: Maldives n Lakshadweep in the Indian Ocean, Seychelles in the western Indian Ocean, Fiji in the South Pacific, n the Canary Islands over in the eastern Atlantic). So I had to choose these specific places because they hve good, matching data from multiple sources. It helps my study. (Wanted massive geographic, geological, and climatic differences so my compound vulnerability score nd governance alignment framework get tested across totally different island styles, law systems, nd administration types instead of jst checking one single boring case).

### Expected Outputs

So I hve specific expected outputs. (1st, I built a settlement level physical exposure dataset for villages across all 5 islands). I also created a validated multi temporal mangrove extent dataset. Then I coded a validated multi decadal coral reef thermal stress dataset using satellite histories. (After that, I mixed these parts to generate a single composite Compound Vulnerability Score nd risk ranking across all 5 islands). This is the core data.

Next comes my political data. So I ran a governance alignment analysis to test the statistical relationship between protected area coverage and verified vulnerability..... (So I am specifically searching for "governance evidence gap" zones where my maps show high verified risk but the governments gave low or no formal protection). So I want to see the gaps.

Then I ran an expansion check. I created a settlement and infrastructure encroachment analysis to test the ecosystem buffer feedback hypothesis..... (I chose to break down these house building trends strictly by protection status to see where lines are failing). Finally, I packaged everything together. I built an interactive geospatial dashboard n an open source codebase tht other students can easily extend to more islands nd completely new ecosystem types.

### Relevance

My project links to real world policy..... (I am talking directly to people working on climate adaptation policy, nature based solutions financing, ecosystem services valuation, nd marine spatial governance)..... I am contributing 2 main things. 1st, I found that treating coastal ecosystem protection as one big mixed category hides totally different risk realities on the ground, so doing strict, ecosystem specific, multi temporal verification matters way more than just believing some lazy uniform decline story..... This is a big eye opener. 2nd, I actually chose to test, instead of jst assuming blindly, whether current conservation governance tools line up with real, verified risk numbers. (By doing this, my dataset gives a fully transferable method so tht anyone can evaluate adaptation finance nd protected area prioritization choices in any coastal or island spot facing compound climate vulnerability). Think this will help a lot.

## Entry 2

### Status

Complete. Covers the initial project structure decisions made once the research question n 5 island study area were locked in, before any data acquisition started.

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## Starting Point

I fixed my research question. (Basically, I want to check if the real physical danger of sea level rise mixes with dying natural shields across 5 islands, n if this varies by ecosystem type). Then I had to make my 1st big decision. So I had to build a project folder structure that actually fits my study needs instead of jst copying some generic template from the internet. This was very necessary.

## Folder Structure: Category Based, Not Island Based

So I had 2 choices before writing any data acquisition code. Could organize by island (meaning a separate Maldives folder, a Seychelles folder, with each place holding all its own data files), or I could organize strictly by data category (a settlements folder, a mangroves folder, with each holding all 5 islands' files together).

Chose the category based plan. (The main reason was my analysis methodology, because I need to compare the exact same variable across all 5 islands at once (like coral bleaching stress or encroachment rates) so an island based structure woudl mean opening 5 separate folders every single time a cross island script runs). It becomes a huge headache. Also, people easily make inconsistent naming mistakes when files are hidden inside separate island folders. Category based keeps every version of a dataset right in one place. My cross island script just runs a simple loop over the island names reading from tht single folder. (It is immediately obvious if any island's file is missing from a category because all 5 files should be sitting right nxt to each other in the directory). This layout keeps my data clean.

The top level structure ended up as:

```
DOUBLE_JEOPARDY/
├── data/
│   ├── settlements/
│   ├── ecosystem_buffers/       (mangroves, coral reefs, protected areas)
│   ├── terrain/                 (elevation, slope)
│   ├── population/
│   ├── cyclone_tracks/
│   ├── boundaries/               (island outlines)
│   ├── mangroves/                (raw multi temporal GMW archives, kept separate
from processed ecosystem_buffers)
│   ├── coral_bleaching/          (NOAA thermal stress time series)
│   └── settlement_encroachment/  (satellite derived built up change data)
├── outputs/
│   └── plots/
├── DJ_Research_Paper.md
├── DJ_Executive_Summary.md
└── DJ_Development_Log.md
```

## One Deliberate Inconsistency

So I made one deliberate inconsistency in my folder setup..... (The raw multi temporal Global Mangrove Watch archives for the 1996, 2010, n 2020 snapshots live inside their own mangroves/ folder, which is completely separate from the clean per island files inside the ecosystem_buffers/ directory). Chose this for practical reasons. The raw GMW shapefiles are absolutely massive since they weigh around 900 to 950 megabytes per year n cover the whole globe. Only need to process them once into tiny per island n per year extracts. (By keeping the giant source archives physically away from the small final files, I easily excluded the heavy raw archives from my Git version control while still tracking the lightweight output files nicely). This saved me from scattering folder level exceptions everywhere.....

### Environment Setup

I reused my old gpie2 conda environment. (I did not want to start fresh because this specific environment was already fixed for a nasty numerical computing issue where a corrupted Intel MKL backend was breaking my scripts). So I solved tht old error by installing NumPy and SciPy with the nomkl OpenBLAS backend instead. Reusing it meant I did not risk facing tht same failure mode again in a clean environment. 2 of my old files carried straight over to this project. I moved auth_sentinelhub.py for the Sentinel Hub OAuth module n my confidential .env file containing the Copernicus Data Space credentials. (So I did this because the API access patterns nd login credentials were 100% identical to wht I need for my new Sentinel-2 imagery work anyway). This saved a lot of setup time.

### Why This Mattered Later

My upfront folder structure paid off beautifully. (Once I started the multi temporal analysis, everything became super easy because every mangrove year file, every coral time series file, nd every island's settlement file sat in a totally predictable, category consistent spot in my directory layout). This choice saved me hours of extra coding later..... My scripts looping across all 5 islands (like the elevation exposure checks, mangrove area comparisons, nd coral stress trends) could be written in a generic way. (Only had to change the island name between iterations instead of writing 5 near duplicate scripts hardcoded to 5 completely different folder setups). This trick kept my development log clean.

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## Entry 3

### Status

So I finished this part completely. (My logs here cover how I pulled down and cleaned the 3 OpenStreetMap derived datasets to build a clear baseline for all local human activity: settlements, tourism infrastructure, and general infrastructure). The files are ready now.

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### Source and Acquisition

I grabbed all 3 datasets from OpenStreetMap. (So I ran specific Overpass Turbo queries filtered exactly to each island's borders, which gave me one raw GeoJSON export per island for each distinct category to create fifteen total files). So I chose OSM because it was practical. It gives me consistent, free data across 5 different national administrative systems: Maldives, India, Seychelles, Fiji, n Spain. (No single government office provides a dataset tht covers all 5 islands in the same format, so I had to use OSM to keep my framework uniform)..... This solved a huge matching problem.

## The Column Bloat Problem

My raw OSM exports had an absurd number of columns. (Usually, I found around 200 to 240 columns in every single file, but most of it was total garbage like place name translations into dozens of languages such as name:ta, name:zh Latn pinyin, or wikipedia:ur, along with Wikidata links and contact info tht had nothing to do with settlement identity, population, or location). I did not want to carry this dead weight around. It made checking files by hand a massive headache.

So, I chose to clean all fifteen files the exact same way. Read them into my script using geopandas nd trimmed down the data to columns that actually mattered..... (For settlements, I kept name, place type, population, nd geometry; for tourism, I chose name, tourism type, amenity type, and geometry; n for infrastructure, I saved name, man made feature type, amenity type, power infrastructure type, nd geometry). Then I saved them as new _clean.gpkg files. (So I left my original raw GeoJSON files untouched so I did not lose any source data). My settlement files dropped from 240 columns down to jst 4 or 5. I did not lose a single row or any useful field during this process.

## Bug: Sea Features Mislabeled as Settlements

So I found a weird bug while checking my clean settlement files. (My bounding boxes for the Maldives nd Lakshadweep data looked way too massive nd extended way outside the actual island borders). 2 specific rows caused this mess in both files. They were labeled "Arabian Sea" n "Laccadive Sea." (These entries were obviously not real human settlements at all, but random OSM features for large water bodies that got accidentally dragged into my Overpass Turbo place type query because online contributors tag things inconsistently). It happens a lot on OSM.

Caught this mistake by doing a quick trick. Calculated the centroid coordinates for every single settlement, sorted everything by longitude, n checked the extreme values at both ends. (The 2 sea name rows stuck out like a sore thumb immediately because their coordinates were sitting far away from the actual settlement clusters). So, I chose to delete both rows directly from my cleaned .gpkg files. (My Maldives file dropped from 998 to 996 rows, nd my Lakshadweep data went from 38 down to 36 rows). This fixed the bug completely without making me re pull anything from OpenStreetMap.

## Bug: Invalid Geometry in Tourism Data

I hit a major code crash. (While I was trying to read the Canary Islands tourism dataset, geopandas.read_file() suddenly threw a bad GEOSException error about an invalid number of points in a LinearRing). The file only had 2 points instead of the 4 pairs needed to make a proper loop. This is a realy common mistake. (People who edit crowd sourced OpenStreetMap data frequently make these broken, half finished polygon lines by mistake).

So I fixed the crash easily. I chose to pass on_invalid="ignore" right inside the geopandas.read_file() command so my script drops unreadable geometries instead of dying. Then I ran an explicit check. (I wanted to count if there were any null geometries left after the read, and I found exactly 2 invalid rows in the Canary tourism file). My rows dropped from 10,877 down to 10,875 valid ones. Made this exact setup a standard step for all my later vector cleaning scripts. (Broken polygon defects can show up anywhere when u use open source OSM files, so adding this check keeps my pipeline safe). This saved my code from breaking again.

## Duplicate and Naming Cleanup

I found a few naming inconsistencies here. (This issue became a recurring pattern throughout my whole project becuase manually re running exports keeps producing near duplicate files under slightly different names). So I had to double check everything. Fiji's protected area adjacent file existed twice in my directory as FIJI_INFRAA_SLR.gpkg with a silly typo extra "A" n as the clean FIJI_INFRA_SLR.gpkg version. They were identical in size. (Chose to delete the typo'd copy directly from my folder to save space). I used this same handling pattern every single time this issue popped up anywhere else. So I always checked the file sizes or row counts 1st to confirm whether the copies actually differ. (Then I deleted the confirmed duplicates immediately rather than leaving ambiguous near identical files sitting around making my folders look messy). This kept my directory clean.

### Final Status of This Phase

| Dataset | Islands Covered | Typical Column Reduction |
|---|---|---|
| Settlements | 5 (2 outlier rows removed from 2 islands) | ~240 → 4–5 |
| Tourism | 5 (2 invalid rows removed from 1 island) | ~200–490 → 4 |
| Infrastructure | 5 | ~200 → 4 |

All fifteen files confirmed clean, correctly bounded, nd free of invalid geometries by the end of this phase. That's the human activity baseline used through the rest of the project, including the encroachment analysis much later.

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## Entry 4

### Status

Finished this part completely. (My log entry here covers how I collected n cleaned the 3 ecosystem buffer datasets tht I need for my main hypothesis testing: mangrove extent, coral reef extent, n protected areas). All 3 sets are done.

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### 2 Different Source Types

My ecosystem buffer data was not like my settlement data. (While I got all settlement, tourism, nd infrastructure layers uniformly from OSM for all 5 islands, the ecosystem buffer data came from 2 completely different source structures depending on the island, which forced me to write 2 separate cleaning pipelines instead of using one shared script). This doubled my processing work.

Found tht Fiji, Lakshadweep, n Seychelles mangrove nd coral data came straight from WCMC's Global Mangrove Watch nd coral reef atlases. (You can identify these files right away because they use a loud uppercase column convention like LAYER_NAME, METADATA_I, NAME, FAMILY, GENUS, SPECIES, DATA_TYPE, GIS_AREA_K, nd REP_AREA_K). But the data for the Maldives n Canary Islands came from OSM. It used the exact same lowercase place, natural, n tourism schema tht I found in my settlement files. (So I had to make sure every mangrove n coral cleaning script could detect which source format a given island was using to apply the correct column selection logic). Had to create a specific WCMC_KEEP_COLUMNS list for the 1st group nd a separate OSM style list for the 2nd group instead of using one universal set. This kept my dataset neat.

## Bug: The Maldives "Coral" File Wasn't Coral Data

So I found a huge data quality problem here. (When I 1st opened the coral_maldives.gpkg file, it had 4,230 rows n 79 columns, but looking at the column names made me realize a bunch of fields had absolutely nothing to do with real coral reefs). The columns were filled with random words like cuisine, spa, restaurant, rooms, payment:visa, payment:mastercard, opening_hours, and parking. This file was not a filtered coral dataset at all. (It was basically an unfiltered general OSM export for the whole Maldives region tht someone had jst named like it was coral specific data). It was a total mess.....

So, I chose to filter it down myself. I used the official OSM reef tagging convention by running natural == "reef" on the table. (This step correctly picked out 2,921 of the 4,230 rows as actual coral reef polygons, meaning the other 1,309 rows were just unrelated tourism, hospitality, and amenity points that got mixed into the same export by mistake). I cleaned it up completely. A reef attribute column with a single consistent value like "coral" across the correctly tagged rows proved to me tht my filter isolated the right feature type. (Doing this filtering step before any column cleanup ensured that my downstream analysis was not quietly ruined by thousands of irrelevant tourism records). This made my data clean.

## Real Ecological Absences, Not Data Gaps

So I got zero results for 2 islands..... (When I was pulling specific ecosystem data, these 2 places came back completely empty, but I chose to investigate n found out they were real absences instead of data download failures). So now I double check every zero result before treating it as an error.

My Lakshadweep data showed zero mangrove features. This makes sense. (Mangroves are very rare in this specific archipelago because it is jst a small coral atoll chain tht lacks the muddy intertidal delta conditions tht mangrove trees actually need to grow). Accepted this as a true fact.....

So I also found zero coral reefs for the Canary Islands. (This fits its subtropical, volcanic, cooler water Atlantic setting perfectly because the island sits way outside the warm tropical shallow water temperature range tht corals need)..... I got independent cross confirmation for this later. Ran a separate Global Mangrove Watch bounding box query for another multi temporal analysis, n it came back totally empty for the Canary Islands even after I deliberately made the search box much wider. (This proved to me that this island's coastal ecology is a different animal from the tropical islands, nd it was not a systematic acquisition problem with my download script). My data looks correct now.

## Protected Areas: Swapping Out a Weak Source

So I hit an issue with protected areas. (For 4 islands, Maldives, Canary Islands, Seychelles, n Fiji, I originally got the protection dataset from OpenStreetMap using the QuickOSM plugin, but on checking I found an implausibly small number of features like just one protected area for Fiji n 4 for Maldives). This data felt completely wrong. Knew these regions hve plenty of officially gazetted marine n terrestrial protected areas, so it was jst bad coverage.

So, I switched my source. I chose to use the World Database on Protected Areas instead because it is the main global dataset from UNEP WCMC. The full WDPA download is massive at abt 4.15 gigabytes nd comes split into 3 sequentially numbered zip archives due to shapefile attribute table size limits. (So I unzipped everything nd filtered each of the 3 polygon shapefiles per island using the bbox parameter inside geopandas.read_file() so my script only read the relevant subset off the disk instead of loading the whole multi gigabyte global file

into my computer memory). This is my standard memory efficient approach. Finally, I combined the results from all 3 archive parts into a single clean file for each island.

My feature counts jumped up a lot. (After I fixed my data source, the Maldives count went from 4 to 93, Fiji jumped from 1 to 126 because I combined 2 sub queries to cover its territory tht straddles the antimeridian, Seychelles grew from 4 to 49, n the Canary Islands went from 258 up to 385 features)..... This proved my old data was bad. I found tht Lakshadweep had zero WDPA registered protected areas..... (This matched my earlier zero result from OpenStreetMap perfectly, giving me solid cross source proof tht this is a real fact abt Lakshadweep instead of some random database glitch). So, I deleted the old bad files n replaced them with the new WDPA data across all 4 islands.

The raw WDPA entries had 34 columns each. (They were carrying a ton of heavy metadata about governance types, ownership types, management authorities, IUCN categories, n verification status tht I did not need at all). Chose to trim them down. So I kept only 9 useful columns: name, designation, designation type, IUCN category, legal status, status year, reported area, nd GIS calculated area. (This is enough information for me to calculate protection type nd size without dragging along useless admin rows tht nothing in my project code actually calls). My final files look much smaller now.

### Final Status of This Phase

| Dataset | Islands With Data | Confirmed Real Absences |
|---|---|---|
| Mangroves | Maldives, Seychelles, Fiji, Canary Islands | Lakshadweep |
| Coral Reefs | Maldives, Fiji, Lakshadweep, Seychelles | Canary Islands |
| Protected Areas (WDPA) | Maldives, Seychelles, Fiji, Canary Islands | Lakshadweep |

This phase set the ecosystem buffer baseline used both for the project's static ecosystem context n, later, as the foundation for the full multi temporal analysis in Entry 6.

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## Entry 5

### Status

I finished this part completely. (My logs here cover the remaining baseline files I need for the project: height nd slope terrain maps, population density, nd old historical cyclone tracks). Everything is processed now.

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### Terrain: Elevation nd Slope

I got my elevation data from Copernicus DEM. (Luckily, the files for all 5 islands arrived in the correct EPSG:4326 coordinate system right from the start, so I did not have to waste time doing any map re projections there). That was an easy win.

But the slope files were annoying. So I found tht 3 islands (Canary, Fiji, n Lakshadweep) arrived in the wrong EPSG:3857 map system instead of my project baseline EPSG:4326. (I had to fix this by writing a script with rasterio's calculate_default_transform n reproject functions using bilinear resampling because tht is the correct way to handle smooth continuous elevation stuff instead of nearest neighbor resampling which is only for category grids like land cover). Seychelles was even worse because it had zero slope files in the folder. (Instead

of crying or downloading new files, I chose to generate the slope layer directly from my already checked Seychelles elevation raster using GDAL's gdaldem slope command in my terminal). This gave me a perfectly matching slope raster without downloading any extra junk. It saved my day.

So I cross checked elevation values manually. (Matched my raster data against real geography maps as a sanity check because I did not want to jst blindly trust random numbers on my screen). My Canary Islands file maxed out at around 3,696 meters. (This lines up nicely with Mount Teide, which is Spain's highest peak sitting at roughly 3,715 meters). The Seychelles data hit a top peak of around 904 meters. (This fits its steep granite mountain geography like Morne Seychellois perfectly). On the other hand, the Maldives and Lakshadweep layers stayed strictly in a super low 0 to 20 meters range. (This matches exactly wht you expect for low lying coral atoll nations on any map). This quick matching test gave me high confidence. My height layers were processed correctly and did not get silently corrupted or shifted.

## Population: Checked Against Real Numbers

So I downloaded the 2020 WorldPop density rasters. (So I managed to get good data for 4 out of my 5 targets: Fiji, Seychelles, Maldives, n the Canary Islands, where I clipped the last one directly from Spain's big national population map). This part went smooth. But Lakshadweep did not get any population data in this phase. (The only online download option I found meant pulling a massive multi gigabyte raster covering the entire country of India, which felt totally stupid nd impractical for the tiny island area I actually needed). So I skipped it for now. Chose to put it as a low priority task for later.

So I validated each raster by myself. (I chose to sum up every single pixel value in the files to calculate a total population estimate, then I checked those sums directly against real world figures instead of jst assuming the data was correct because the code ran fine). My verification showed good numbers. The Canary Islands file gave me 2.09 million against the actual 2.2 million count. (Fiji gave me around 896,000 instead of 900,000, Seychelles calculated to 98,000 against the real 100,000, nd the Maldives raster hit 540,000 which is exactly inside its known real range). So I treated this step as highly essential. A file processing cleanly does not mean the data inside is correct.

Then my code hit a wall. (My biggest population file for Fiji crashed the script with a rasterio.errors.RasterioIOError because it could not allocate 884,736,000 bytes during a full array read). I had to change my approach..... I fixed this issue by reading the raster in small tiled blocks using rasterio's block_windows iterator instead of opening everything at once. (Accumulated the sum, minimum, nd maximum counts incrementally across the blocks, which is the exact same memory efficient pattern I use for all heavy files in this folder). This trick saved my system memory. Phew.

## Cyclone Tracks: IBTrACS n a Wind Speed Consolidation Problem

I got historical cyclone track data from IBTrACS. (This is NOAA's standard global cyclone dataset that mixes reports from more than a dozen different meteorological offices). It was a very big file. I chose to filter down the full global shapefile per island using the exact same bounding box geopandas.read_file(bbox=...) approach tht I already used for my WDPA files. This method saved my computer from crashing.

4 out of my 5 islands returned real storm data. (I found that Fiji had 143 tracks, Lakshadweep had 4, Maldives had 5, and Seychelles had 2 entries in the final count). These numbers matched my expectations. But the Canary Islands file came back with absolutely zero rows. (Checked this specific place twice under 2 separately named filtering attempts in my script, but it is a true result because the Canary Islands sit entirely outside the typical Atlantic tropical cyclone formation n travel corridor). I logged it as a real geographical finding.

So I found another bad data quality issue here. (The wind speed column I wanted to use, called WMO_WIND, was completely blank for most of my records since it only had data for 1 out of 5 storms in the Maldives nd 31 out of 143 for Fiji). This was a total block. But I noticed tht IBTrACS separately logs wind estimates from different local national agencies like USA_WIND, REU_WIND, BOM_WIND, TOK_WIND, CMA_WIND, n HKO_WIND. So, I chose to use pandas' bfill(axis=1) trick across a prioritized list of these agency columns. (This smart code script automatically grabs the very 1st available non null wind value for each storm row no matter which country's office tracked it originally). Managed to improve Fiji's useful wind speed data from 31 up to 87 out of 143 rows. My data counts for the other 3 small islands (Maldives, Seychelles, nd Lakshadweep) stayed low, but tht is jst because those places hve a real natural scarcity of storms rather than a bug in my coding consolidation method. (Confirmed this fact by running the exact same multi agency check script against those islands nd found absolutely nothing extra beyond wht my 1st 4 agency check caught).

So I saved this new consolidated wind speed field into a clean file. (So I trimmed down the huge raw IBTrACS table which had over 150 useless columns down to jst a small set containing storm ID, season, name, timestamp, nature, position, consolidated wind speed, n distance to land). I saved everything as a new _final.gpkg file for each island. This new file replaced the bulky raw file entirely for my next steps.

### Final Status of This Phase

| Dataset | Islands With Full Coverage | Notes |
|---|---|---|
| Elevation | All 5 | Cross checked against known peak elevations |
| Slope | All 5 | 3 islands reprojected; Seychelles generated from elevation |
| Population | 4 of 5 | Lakshadweep deferred; all 4 checked against real world totals |
| Cyclone Tracks | 4 of 5 | Canary Islands confirmed absent; wind speed consolidated across agencies |

This phase closed out the full baseline dataset inventory needed before starting the project's core multi temporal nd cross dataset analyses, covered nxt.

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## Entry 6

### Status

So I finished this main chunk completely. (My logs here cover the core analytical part of the project where I calculate actual multi temporal ecosystem trend files instead of jst relying on lazy, single before nd after snapshots to measure physical danger). This is the biggest part.

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### Why Multi Temporal

My old data from Entry 4 only showed a single snapshot in time. (So I realized tht if I really want to test whether mangroves or coral reefs are rotting away, I absolutely need dataset layers recorded across multiple, well separated years). A single comparison is a bad idea. (It is too easy for a simple before n after check to look fake because of random noise or an unusual reference starting period). My data needs to be bulletproof.

A trend finding tht holds true across several independent time blocks on a timeline is simply a lot more convincing to examiners.

## Mangrove Time Series: 3 Points

I used the Global Mangrove Watch archive..... (The GMW archive gives you pre processed, ready to use annual mangrove extent snapshots so I did not hve to waste my hours sitting n calculating the boundary limits from raw satellite pictures by hand). It saved my energy. I chose 3 specific time points for my project: 1996 which is the earliest baseline available, 2010 for the exact midpoint, nd 2020 as the most recent single year data block. (This covers a proper 24-year span with a clear check right in the middle because a single big 24-year jump completely hides whether the jungle loss was gradual or jst happened suddenly in a tiny window). I wanted to see the full timeline clearly.

Noticed a small trick inside the GMW folder. (The database provides files both as single year snapshots like gmw_v3_2020_vec.zip n as change detection vectors spanning a date range like gmw_v3_f1996_t2020_vec.zip where f means from nd t means to). So I had to make a choice. I chose to independently derive the total area change by my own scripts using the 3 separately verified snapshots. (So I did not want to blindly lean on GMW's own pre computed change detection algorithm, so I downloaded n ran the 3 single year files instead of touching their mixed pre computed change file). This way every step of the area calculation is in my own scripts.

I filtered each global GMW shapefile. (The raw files are massive at around 894 to 948 megabytes each, so I used the exact same bbox based approach from my WDPA and IBTrACS scripts to avoid loading the whole global file into my laptop memory). This kept my machine from slowing down. So I generated clean per island nd per year mangrove extent files from this step. (Only did this for the 4 islands where mangrove trees actually grow because my data check in Entry 4 already confirmed tht Lakshadweep has zero mangroves). My folders are fully organized now.

**Bug: Feature Count Isn't a Valid Area Proxy**

So I ran an early data check. (I used raw polygon feature counts across the 3 time points as a quick way to see changes, n it looked like Fiji had a massive drop because it went from 5,608 features in 1996 down to 4,810 in 2020 which looks like a 14.2% loss). Did not trust this number. I knew that feature counts nd actual mapped areas are completely different things on a map. (For example, one large continuous mangrove forest can show up as a single polygon in one year, but it can easily break into several small adjacent polygons in another year purely because of how the satellite classification algorithm decides to slice that specific imagery). There might be zero real area change. Had to write a better test.

I tested this issue directly. (So I reprojected all 3 years for each of the 3 mangrove islands into an equal area CRS called EPSG:6933 because it is appropriate for real area calculations unlike standard geographic coordinates which badly distort shapes depending on ur latitude). Then I summed the total area in square kilometers instead of using polygon counts. So I got a totally different, more reliable result. (Fiji's mangrove area was basically stable across all 3 points sitting at 485.72 square kilometers in 1996, then 487.97 in 2010, n 488.41 in 2020 which means a small net increase of just 0.6%). My calculations for the Maldives nd Seychelles also showed less than 0.01 square kilometers of real change over the full 24-year span. So the fake decline from feature counts was jst a segmentation artifact between different years' imagery. The actual mangrove extent held essentially stable across all 3 tested islands over nearly a quarter century, which directly changed how I framed my hypothesis testing later in the project log. This step saved me from making a big mistake.

## Coral Reef Condition: A Thermal Stress Time Series

Coral degradation shows up as bleaching. (Basically, it is a health nd condition change coming from heavy thermal stress instead of a simple drop in the mapped physical boundary line, so I absolutely needed a totally different data source nd metric than my mangrove tracking setup). My 1st plan failed. So I considered using the Allen Coral Atlas 1st, but it turned out to be jst a single snapshot composite built from 2018 to 2020 pictures instead of a real multi year timeline layer. It did not fit my trend requirement.

So, I chose a better tool. (Found tht NOAA Coral Reef Watch's Degree Heating Week product was the perfect match because it gives a continuous daily satellite derived measure of accumulated heat stress from 1985 up to the present day). This dataset tracks everything nicely. (I konw that values above 4°C weeks cause major coral bleaching n numbers above 8°C weeks mean severe bleaching n dead corals)..... This metric measures the actual biological stress driver directly. It makes complete sense because hard coral reef structures can physically stand there in the ocean for years after a bad heating event even when the living animals inside are already dead.

**Bug: Access Denied, Then the Wrong Variable Name**

I accessed NOAA's ERDDAP server. (I sent constructed URL requests directly to their system instead of clicking links to download files manually, which matches the API based coding patterns I use everywhere else in my project folder). My 1st code attempt failed. The server blocked me with a nasty HTTP 403 Forbidden error because it was rejecting Python's default requests user agent string as an automated bot script. I fixed this issue quickly. (Explicitly set a real browser style User Agent header line using Mozilla/5.0 (Windows NT 10.0; Win64; x64) inside my script, after which the server finally started accepting my connections). It returned a 500 error instead of a 403. My access was granted, but I had a different problem left to solve.

The 500 code gave me a clear hint. (Teh exact error string destinationVariableName=CRW_DHW wasn't found in datasetID=noaacrwdhwDaily meant I had written down the wrong variable name inside my script request block)..... So I had to look it up..... Checked the dataset's actual online metadata webpage directly with my browser and found the correct column name, degree_heating_week. (I swapped this proper string right back into my Python code, n it fixed my API request completely). My data started downloading nicely.

**Bug: A Coordinate That Landed on Solid Ground**

4 islands worked perfectly right away. (Got valid, non null DHW series numbers for them using representative offshore coordinates that I picked out from my map grid). But my Canary Islands data failed. The 1st coordinate point I chose came back entirely empty because 305 out of 305 values were totally null. (I traced this bug nd found my coordinate was falling exactly on the landmass or way too close to a beach pixel in their 5km resolution grid, which happens easily because the Canary Islands are very small). It was a frustrating block.

Chose to test 3 alternative offshore coordinates instead of wasting time. (I ran a quick single month request for each new candidate point instead of lazily repeating a full 24-year download for every single location). This smart shortcut worked. One point sitting just south of Gran Canaria returned a valid value. (I immediately used this working coordinate to re pull the full Canary Islands time series successfully without any errors). My coral data is fully complete now.

**DHW Trend Result**

I compared 2 different time blocks. (So I took an early reference period from 1996 to 2000 n calculated its average against a recent period spanning 2016 to 2020 to check for changes)..... My results showed a clear pattern. So I found tht 4 islands had rising thermal stress numbers..... (My calculations showed the Maldives went up by +0.17°C weeks, Fiji hit +0.10, Lakshadweep got +0.08, and the Seychelles jumped up by a massive +0.68 with a single max reading of 10.47°C weeks which sits right in the dangerous range for severe bleaching n coral death). This Seychelles number is very scary.

But my Canary Islands data looked totally different. (It actually showed a slight decrease of −0.05, making it the only island out of my full group of 5 that was not trending upward on the heat charts)..... I checked why this happened. (I realized this fits perfectly with its distinct Atlantic climate regime because it operates completely differently compared to the other 4 Indian Ocean n Pacific islands in my folder). My data mapping looks ready now.

## Physical Exposure: Settlement Level Sea Level Rise Risk

So I calculated the physical exposure numbers nxt. (I chose to sample my elevation raster right at each settlement's center point using rasterio's sample method across all 5 islands so I could find the percentage of villages sitting at or below the one meter sea level rise mark). This gave me a very clear picture. (My data showed a massive geographical spread where 99.1% of Maldives settlements are in high danger, followed by 78.3% for Seychelles, 77.8% for Lakshadweep, 32.0% for Fiji, nd just 12.1% for the Canary Islands). The gap between them is huge. (These results directly show the big difference between flat coral atoll islands nd tall volcanic mountain grounds tht I already saw during my height checks back in Entry 5). My code script ran perfectly.

## Compound Vulnerability Score

Mixed the 2 risk signals. (So I took my independently checked physical sea level rise exposure numbers nd my coral thermal stress trend data, scaled each of them onto a basic 0–1 range using min max across the 5 islands, n simply averaged them with equal weighting). This was a major calculation step. So I chose to leave the mangrove trend out of this formula entirely. (My long term data analysis did not find any real area loss for mangroves, so there was basically no measurable drop to weight in anyway). This made my logic very clear.

My final list ranking came out. I found that Seychelles scored highest at 0.880, followed by Maldives at 0.651, Lakshadweep at 0.467, Fiji at 0.217, and the Canary Islands sitting at 0.000 as the reference minimum by construction..... This proves my project's main argument. (See, the Maldives has a much higher physical risk by itself with 99.1% versus Seychelles' 78.3%, but Seychelles wins the highest total danger spot once u actually mix in the bad coral reef dying trends). Looking at water lines alone is a trap. (If I only looked at physical exposure, I would hve picked the completely wrong island as the most dangerous zone). My composite approach solved this issue.

## Final Status of This Phase

| Analysis | Result |
|---|---|
| Mangrove extent (1996/2010/2020) | Stable across all 3 tested islands; no decline detected |
| Coral thermal stress (1996–2020) | Increasing in 4 of 5 islands; Seychelles most severe |
| Physical SLR exposure | Wide variation, 12.1%–99.1% of settlements at risk |

| Analysis | Result |
| --- | --- |
| Compound Vulnerability Score | Seychelles ranked highest; Canary Islands lowest |

# Entry 7

## Status

Finished the 2 final analytical pieces..... (For this last step, I ran statistical tests to see if official protected area lines actually match up with my verified risk scores, and then I checked whether local villages are encroaching right into the fragile ecosystem buffer zones over multiple decades). So I also put together a complete synthesis of the whole project log. Everything is fully done now.

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## Adding a Governance Dimension

I finished my main risk analysis. (I added an extra phase here to connect the results to policy). I had a plan. Instead of treating protected area boundaries as jst some unexamined backdrop maps, I formulated a specific testable hypothesis. (So I think tht official protection marks like WDPA coverage should go up in places where my Compound Vulnerability Score from Entry 6 is high, if the governments are actually listening to data). So I'll test tht.

## Bug: Comparing Land Area to Marine EEZs

My 1st data test failed completely..... (Tried to calculate a simple protected area to land area ratio, but my script gave me a totally nonsense result of 1005.69 for the Seychelles which basically means the protected zone size went over the actual island land size by more than a thousand times). It was a ridiculous output. Traced this bug down into the code. (Found tht the massive WDPA dataset accidentally included the entire Seychelles Exclusive Economic Zone because my simple bounding box filter from Entry 4 swept up over a million square kilometers of open ocean along with the real land based parks). That marine area is huge. Comparing tht gigantic sea number against a tiny island land area denominator was nvr going to give me any real, readable results.

So I fixed this error quickly. (So I restricted my protected area calculation to a strict 10km coastal buffer around each island's boundary instead of using the full captured WDPA extent because a settlement's actual exposure is tied to nearby coastal protection nd not to some random offshore marine park hundreds of kilometers away). This made much more sense. Implemented this by buffering each island's boundary geometry by 10km after reprojecting to an equal area CRS. (Then I intersected tht fresh buffer zone with each island's WDPA polygons before summing the total area). Much better ratios came out..... Seychelles dropped from 1005.69 down to a clean 11.32. (It is still the highest value of my 5 islands but it is no longer totally absurd, while my Maldives score hit 3.90, Canary Islands got 2.32, Fiji sat at 0.42, n Lakshadweep landed on 0.00 which matches its verified zero coverage from Entry 4). My numbers are fully sorted now.....

## Governance Alignment Result

So I ran a Pearson correlation test next. (So I took my Compound Vulnerability Score for each island and checked it against its coastal buffer WDPA ratio using the scipy.stats.pearsonr function in my Python script). The output gave me r=0.727. This means I found a moderately strong positive relationship. (See, my data shows tht Seychelles has the highest vulnerability nd also carries the highest protection ratio, which looks like

a good pattern). But my p value came out to 0.164. (This means my test missed the conventional 0.05 statistical threshold completely so it is not 100% solid proof). I logged this straight as suggestive rather than confirmatory. The direction definitely looks like risk responsive governance. But the main headache is tht 5 islands jst do not give enough statistical power to be fully confident abt it, which is the exact same small sample size limitation tht ruined a comparably small country sample earlier in my old project work. I cannot change this reality.

**Bug: A NoData Artifact Hiding Inside My Canary Islands Number**

Went back n double checked something tht had been bugging me a little. (My Canary Islands physical exposure number of 12.1% from Entry 6 jst felt off compared to how flat nd low lying its elevation profile looked in my earlier height checks, so I decided to open up the raw per point raster sample results instead of jst trusting the summary percentage). Good thing I did tht. (Dug through the point by point output n found tht 649 out of my full 5,483 raster sample points for the Canary Islands were actually NoData values sitting right at the coastline edge of my raster grid, but my earlier script was silently counting a chunk of those NoData points as if they were real, valid at risk settlement points instead of throwing them out first). Tht is a real bug, not jst noise in the numbers.

Fixed it by properly excluding those 649 NoData points before doing the percentage math. (Once I recalculated using only the properly valid raster samples, the Canary Islands physical exposure number dropped all the way down from 12.1% to a much more realistic 0.3%, which actually fits its flat volcanic coastline way better than tht old inflated figure ever did)..... Phew. Because this feeds directly into my Compound Vulnerability Score formula from earlier in this entry, I had to rerun tht whole calculation too. My corrected scores came out as Seychelles at 0.895, Maldives at 0.651, Lakshadweep at 0.481, nd Fiji at 0.263, with the Canary Islands still sitting at 0.000 as the reference minimum (the ranking order stays exactly the same, Seychelles is still on top, jst the individual numbers shifted a little bit). This also meant redoing my Pearson correlation test right above since it leans on these same vulnerability scores. (My corrected governance alignment result came out as r=0.718 nd p=0.172, which is basically the same moderately strong, not quite statistically significant pattern as before, jst slightly adjusted). Nice tht my overall story n ranking didn't change at all, but I'm really glad I caught this before locking anything in as final 😃

## Settlement Encroachment: Satellite Derived Built Up Area Change

Coded the last analytical part nxt. (Chose to test if new houses n infrastructure expansions are crowding right inside or close to the fragile ecosystem buffer zones over a long time timeline by using the Normalized Difference Built up Index). This is called NDBI for short. (So I built it the exact same way people make standard NDVI but I tuned my code parameters to detect built up urban surfaces instead of green trees nd grass by calculating values directly from Sentinel-2 shortwave infrared and near infrared bands). It ran nicely. Applied this index math strictly to the 3 islands where mangroves are actually growing: Maldives, Seychelles, nd Fiji. (My script compared an old 2016 baseline satellite dataset directly against a recent 2024 time period to map out the new construction expansion). This gives me the final building numbers.

**Bug: Empty Results, No Error**

My 1st data pull failed quietly. (So I used a full calendar year aggregation window called P1Y for each island, which returned an HTTP 200 success code but gave me a totally empty data array like "data": [] nd a "geometryPixelCount": null field for every single request across all 3 islands nd both periods). This was an

annoying silent failure. (So I had to read the raw JSON text directly to catch this bug because I couldn't just blindly trust the green request success status alone). It was tricky.....

Made 2 changes to fix it. Switched the aggregation window length from a full year down to 6 months using P6M. (I chose this because of a lesson from my older $NO_2$ nd NDVI data work where certain Sentinel Hub interval lengths behave inconsistently depending on your specific data type). Then I did one more step. I added an explicit maxCloudCoverage: 30 filter line to my request script. (Persistent empty values across a full year for tropical islands usually mean heavy cloud cover is blocking any usable clear sky satellite observations). Both edits worked perfectly together. All 6 of my requests (meaning 3 islands multiplied by 2 time periods) came back fully populated with numbers.

**Encroachment Result**

Compared the average NDBI values. (Looked closely at the data between my 2 selected time blocks and found tht both Maldives went up by +0.0461 n Seychelles grew by +0.1135 to show a clear increase matching real house nd infrastructure expansion right near the ecosystem buffer zones between 2016 nd 2024)..... Seychelles showed the largest growth jump. (My data reveals it actually crossed completely from a negative average in 2016 which means vegetation dominated lands straight into a positive value in 2024 showing it is now built up dominated ground). This is a massive change. But Fiji showed basically zero shift sitting at −0.0008 on my calculator. (This dead flat trend lines up perfectly with the overall stability tht I already saw earlier when checking its mangrove extent dataset). My numbers match across different metrics.

## Final Synthesis

Seychelles kept popping up everywhere. (Across every analytical piece I coded, this specific place showed the absolute strongest proof of a compounding, self reinforcing vulnerability pattern because it carried the highest Compound Vulnerability Score driven by the worst coral thermal stress trends). My data points in one clear direction. It also has the highest coastal protected area ratio. (Even though I could not statistically confirm this governance response with my small sample size, the policy direction looks correct). It also gave me the biggest house building encroachment signal.

Fiji was the opposite story. (My data shows it stayed completely stable across every single temporal measure over the decades with flat mangrove boundaries, the smallest coral heat stress increase among all working islands, n zero infrastructure encroachment). This was an amazing comparison. It serves as a rock steady low change reference case right against Seychelles' crazy high change profile in my sheets. My full synthesis is complete. My mangrove decline hypothesis completely failed. (I tested H2 the same way as everything else, nd the data did not support it on any of the 3 islands with mangroves). So I found no real jungle loss. (My data clearly shows that nature buffer degradation is not uniform across different ecosystem types at all because coral reefs in my 5 island sample are measurably more vulnerable to heat stress than mangroves over this multi decadal timeline). My final log sheets are sorted.

## Complete Project Data Inventory

| Category | Coverage |
|---|---|
| Settlements, Tourism, Infrastructure | 5 islands each |
| Mangroves (multi temporal, 3 time points) | 3 islands (Lakshadweep and Canary Islands confirmed absent) |

| Category | Coverage |
|---|---|
| Coral Reefs (extent) | 4 islands (Canary Islands confirmed absent) |
| Coral Thermal Stress (24-year time series) | 5 islands |
| Protected Areas (WDPA) | 4 islands (Lakshadweep confirmed absent) |
| Elevation, Slope | 5 islands each |
| Population | 4 islands (Lakshadweep deferred) |
| Cyclone Tracks | 4 islands (Canary Islands confirmed absent) |
| Settlement Encroachment (NDBI) | 3 islands (mangrove present islands only) |

Investigated every single zero result. (Whenever I hit an unexpected finding along the way, like Lakshadweep's completely absent mangroves, the Canary Islands' absent corals and cyclones, or my mangrove decline hypothesis not holding up, I chose to dig deep n confirmed them as real geographical facts or traced them back to a specific, fixable data cause). Where a hypothesis didnt hold, I wrote it down as a result.

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# Entry 8

## Status

Added a final part here. (So I chose to check cyclone driven satellite damage detection to get supporting evidence for my risk numbers, but I hit a very annoying data availability constraint while working on the script). It was a surprise block. Wanted to see if the satellite imagery could prove the storm damage on the ground immediately. (But my data search showed tht finding clean, cloud free images right after a major cyclone is nearly impossible for these tiny tropical islands). This limited my calculation steps. So I logged this asset gap as a real project constraint for later.

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## Context n Why It Got Added

My core analysis centers on sea level rise and ecosystem buffer degradation. (Made a deliberate scope call to treat cyclone activity as a background factor rather than a primary pillar because data from an earlier, differently framed version of this island nations project showed meaningful cyclone intensity variation only for Fiji, with the other 4 islands' historical storm records being too weak and low intensity to detect any real damage signal on a satellite map). This earlier version was originally built around cyclone impacts specifically. So I changed my mind.

So I did not want to throw away tht earlier cyclone related work once my project's framing shifted toward water lines nd coral death. (I chose to keep it as a smaller, clearly scoped supporting addition inside my folders instead of messing up nd re expanding my core analytical structure). This saved a lot of code. 2 Sentinel-2 NDVI before and after comparisons from tht older project phase were still sitting right inside my data folder..... (These maps cover Fiji's 2 strongest historically recorded cyclones, Winston in 2016 n Yasa in 2020, n they stayed completely unaffected by my later project rename). So I am glad I saved them 😃

## Trying to Extend This to the Other 4 Islands

I tried an extra test 1st. (Before I finally settled on Fiji as my only island with clear cyclone damage evidence, I tried extending the exact same satellite damage detection approach to the Maldives, Seychelles, n Lakshadweep using each island's single strongest recorded storm from the IBTrACS database tht I already set up in Entry 5). This took some extra time. 1st, I had to pin down each island's peak intensity event name nd its exact calendar date. (I needed to do this step because building a proper before nd after satellite imagery window completely depends on knowing the exact day when the storm actually hit the beach). This step was very necessary for my scripts.

## Ran Into a Satellite Record Start Date Wall

I hit a major database wall while writing my acquisition script. (Found out a huge fact tht nvr came up before this step: Sentinel-2, which is the exact satellite I use for all my green NDVI maps in this project folder, only launched in the year 2015, meaning zero pictures exist before tht date). It was a total shocker. Then I checked the historical dates for my 3 candidate islands' strongest recorded cyclones against this satellite launch cutoff line. (My data matching showed tht all 3 storms happened way before Sentinel-2 was even built: the Maldives' biggest cyclone hit back in 2006, Seychelles faced its worst one in 1996, nd Lakshadweep had its massive peak storm in 1997)..... This made things impossible. No valid "before" satellite imagery could evr exist for any of these islands no matter how smartly I chose to code the API requests. The satellite hardware was simply not flying around in space yet during those old years. This data scarcity completely killed tht line of analysis.

I built this check directly into my script. (I chose to add a strict pre condition logic line tht checks the cyclone year against a 2016 cutoff rule before my code attempts to run any image download requests). This saved me from extra debugging work. (Instead of letting my code run blindly and return an uninformative empty response that would need separate manual diagnosis later, the code stops early if the dates are too old). So I use this exact upfront constraint pattern everywhere else in my project folder. Running my code confirmed everything worked. (My script skipped all 3 islands immediately n printed out an explicit, clear reason on my terminal screen for each skipped folder). It handled the gaps perfectly.

## How This Got Reported

I didnt swap in a later, weaker cyclone to get a result. It is a structural limit of satellite based verification..... (Any remote sensing method u try is strictly bounded by when its satellite hardware platform actually started operating in space, n events before tht date cannot be verified by that method no matter how smartly u configure the API request parameters). So I wrote it down as a scope limit for this one supplementary analysis. (It does not touch my project's core sea level rise or ecosystem degradation findings because those calculations do not depend on Sentinel-2's coverage window the same way at all). My primary results stay fully safe.

## Final Cyclone Damage Proxy Status

| Island | Cyclone Damage Evidence | Reason |
|---|---|---|
| Fiji | Available (Winston 2016, Yasa 2020) | Both events post date Sentinel-2 availability |
| Maldives | Not available | Strongest recorded cyclone (2006) predates Sentinel-2 |

| Island | Cyclone Damage Evidence | Reason |
|---|---|---|
| Seychelles | Not available | Strongest recorded cyclone (1996) predates Sentinel-2 |
| Lakshadweep | Not available | Strongest recorded cyclone (1997) predates Sentinel-2 |
| Canary Islands | Not applicable | No cyclone events recorded in this region (Entry 5) |

Analyzed the 2 Fiji results nxt..... (My numbers showed a clear NDVI decline of −0.0246 after Cyclone Winston which hit at 150 knots as a Category 5 equivalent storm). For Yasa I originally reported zero clear decline, +0.0049, but on a later check I realised tht number came from pairing the earliest pre cyclone reading I had (September, more than 3 months before Yasa even made landfall) with the earliest post reading, instead of the pre/post windows actually closest to Yasa's 17 Dec date. Redone with the closest pre window (late Oct–Nov) nd closest post window (Jan), Yasa actually shows a decline of about −0.0275, bigger than Winston's, not smaller. So the "satellite detectable vegetation damage only shows up at extreme storm intensity" line doesn't hold up nd I'm dropping it. (Also doesn't match the broader ecosystem stability I saw in Fiji's mangrove dataset the way I originally thought it did: tht mangrove finding still stands on its own, it just isn't corroborated by this cyclone comparison anymore). Keeping both numbers here as supplementary Fiji only evidence same as before, still not extrapolating to the other 4 islands, just correcting wht the comparison itself actually shows.

This specific case was a bit different. (Most of my earlier debugging work was abt verifying data was correct once it had already been acquired by checking bounding boxes, validating population totals, or confirming real ecological absences). This time I faced a unique challenge..... I needed to catch a structural limit of the data acquisition method itself. (I had to check the satellite's fixed launch date before even attempting the acquisition loop instead of only finding out through an empty result after running the script). This quick trick saved my processing time.

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# Entry 9

## Status

Went back through my data quality checklist again for one more check..... (Just wanted to be sure before calling this whole section done). Found something I missed the 1st time. Turns out the Canary Islands zero elevation fix from Entry 7 wasnt the full story after all. Fiji had the exact same problem sitting there this whole time. Nobody had checked it yet.

## Fiji Never Got the Same Check Canary Got

So back in Entry 7 I caught tht Canary's zero elevation DEM readings were a NoData artifact nd not real terrain, nd fixed it (12.1% → 0.3%). (But I never actually asked if any other island had this same problem hiding in it). Went back nd counted it properly this time. 649 of Canary's 5,483 points read exactly 0.0m, tht's 11.8%, the number I already caught before. Fiji has 414 of 1,323 points reading exactly 0.0m too. Tht's 31.3%. An even bigger spike than Canary's own number. Same kind of volcanic, mountainous terrain where real settlements

sitting at sea level jst dont make geological sense. I hadnt touched Fiji's number at all before this. So its reported exposure of 32.0% was carrying this same artifact the whole time nd I never noticed.

Went nd checked wht kind of settlement points these actually were before assuming anything. (Islet, village, hamlet, archipelago, a real mix of place types, not jst big place label centroids sitting over open water). So it's not an easy "the label point fell in the ocean" explanation. A real DEM shouldnt return a literal, bit exact 0.0 for 31% of points on a mountainous island. A real elevation distribution near sea level would show a gradient, 0.3m, 0.8m, 1.4m, nd so on, not a spike sitting exactly on zero. Also found tht map1_slr_fiji.py, the script tht makes the standalone Fiji map figure, was ALREADY excluding Fiji's zero readings on its own. Completely independent of the main pipeline, which never did this. So 2 parts of my own project were quietly disagreeing with each other this whole time nd I never noticed either.

Fixed it by adding fiji to the same nodata exclusion set already used for canary. Did this across precompute_elevations.py, dashboard/precompute_elevations.py, slr_exposure_analysis.py, nd export_settlement_elevations_full.py. Corrected Fiji's settlement based exposure from 32.0% down to 1.0% (909 valid points left after dropping the 414 NoData ones, only 9 truly at risk). This is a much bigger correction than Canary's, proportionally speaking. But Fiji was already my 2nd lowest exposure island before this fix, so it doesnt flip anything in the ranking order. Jst makes the "volcanic islands have way less exposure than atolls" story even more lopsided than wht I'd already reported.

One open point. Maldives (99.0% of points exactly 0.0m) nd Lakshadweep (77.8%) show this SAME zero heavy pattern. I did NOT apply the same fix to them. (The difference is tht Maldives nd Lakshadweep are real low lying coral atolls). Near sea level elevation across most of the land is the expected geology there, not some anomaly the way it is on a volcanic island. So the same statistical red flag doesnt carry the same weight for those 2. But I also cant say for sure none of their zero readings are DEM artifacts either. Havent done a pixel level QGIS check on either of them the way I did for Canary. Leaving this open for now.

## This Changed the Compound Score nd Governance Test Too

Since Fiji's exposure number feeds straight into the Compound Vulnerability Score, had to rerun tht too. Fiji's score dropped from 0.263 to 0.106. Ranking order didnt change though (Seychelles still highest, Canary still the reference minimum at 0.000). Fiji jst moved further down within where it already was. Reran the weighting sensitivity sweep too nd confirmed the Seychelles vs Maldives crossover point (abt 76.8% physical exposure weighting) is completely unaffected. Fiji was never the min or max input either the physical exposure or coral normalization got scaled against anyway.

## Fixed the Lakshadweep Zero Hiding Inside My Governance Correlation

While I was back in tht part of the pipeline, caught another thing tht'd been bugging me. Lakshadweep's WDPA coastal ratio is reported as 0.00 because I flat out dont hve a protected area dataset for it. (I already documented tht correctly as missing data back in Entry 6). But I was still feeding tht placeholder zero into the Pearson correlation like it was a real 5th data point. Tht's not "reporting a caveat," tht's letting a made up number quietly bias an already tiny 5 point correlation. Fixed governance_correlation_test.py to drop Lakshadweep from the correlation itself (n=4) nd report its vulnerability score separately, unpaired against any WDPA number. New result: r=0.862, p=0.138 (up from r=0.718, p=0.172 with the old n=5 version tht still counted Lakshadweep's fake zero). Still not statistically significant at this sample size. Still the same "suggestive, not confirmed" story. Jst computed the right way this time.

## Switched the Coral Trend Test to the Seasonal Mann-Kendall Version

Also realized my Mann-Kendall trend test on the 24-year DHW series was using the plain, original version, run straight on raw monthly values. DHW is seasonal by nature though..... near zero most months, then spiking during the warm season. So the plain test's independence assumption jst doesnt hold for data shaped like tht. Switched coral_trend_test.py over to pymannkendall's seasonal_test(period=12), which compares same calendar month values across years instead. New p values came out as: Maldives 0.0046 (was 0.011), Seychelles 0.0069 (was 0.025), Fiji 0.2437 (was 0.184), Lakshadweep 0.3725 (was 0.386), Canary 0.6196 (was 0.641). Nothing flips. Maldives nd Seychelles are still the only 2 significant, everything else still isnt. The seasonal test is the right one for this data, nd the result didnt change.

## Cleaned Up Some Wording nd Some Dead Code

- The NDBI "settlement encroachment" section (paper §3.8/4.7, dashboard) was worded like it measured change right near mangrove/reef buffer zones. It doesnt. It's jst a single mean NDBI value over a broad bounding box around each island's main settled area. Reworded it throughout to say wht it actually measures instead of wht it sounds like it measures. The numbers themselves (Fiji −0.0008, Maldives +0.0461, Seychelles +0.1135) didnt change. Jst the framing around them.
- normalize_wdpa.py nd compute_wdpa_area.py, these are the OLD WDPA scripts from before Entry 6 tht divide marine+terrestrial WDPA extent by land area (the ones tht produced Seychelles' impossible 1005.69 ratio). They're already unused since governance_correlation_test.py reads wdpa_coastal_normalized.csv, not wdpa_normalized.csv. But I'd left them sitting in the repo with zero indication they were superseded, which jst reads as confusing dead code to anyone going through my project folder. Added a clear header to both explaining they're kept only as a record of tht old bug, not part of the live analysis.
- Fixed the mangrove row in my own data inventory table from Entry 8. It said "4 islands (Lakshadweep confirmed absent)" for mangroves, copy pasted straight from the WDPA row above it nd never actually fixed. Mangroves were only ever tested for 3 islands (Maldives, Seychelles, Fiji). Canary Islands has none either, same as Lakshadweep. The paper nd the actual code (mangrove_3point_comparison.py) were always correct on this. It was jst this one summary row in my own log tht had the leftover typo.
- Also went through nd fixed "5 island nations" → "5 islands" everywhere in this log (including my own title above). Lakshadweep is a Union Territory of India nd the Canary Islands are an autonomous community of Spain. Neither one is a nation. The paper, README, nd executive summary already said "5 islands" correctly. This log was the only place still calling them nations.
- fix_map_resize.py (both the root copy nd the dashboard copy) was injecting a JS snippet checking `typeof map !== 'undefined' && map.updateSize()` into the folium webmap HTML files. Neither half of tht check was ever going to work. Folium names its map variable with a random hash (e.g. `map_bbe0fe6c...`), not literally `map`. Nd Leaflet's resize method is `invalidateSize()`, not `updateSize()` (thts an OpenLayers method). So this "fix" from wherever I originally wrote it was dead code the whole time, never actually resizing anything. Rewrote it to scan `window` for any Leaflet `L.Map` instance nd call `invalidateSize()` on whichever ones it finds, no matter the variable name, nd reran it against all 10 committed webmap files to apply the corrected version.
- Also fixed population_weighted_exposure.py to mask against each island's real boundary polygon instead of jst a rectangular bounding box. A bbox can pull in open water pixels for an irregular archipelago (Fiji, Seychelles) tht a real island boundary wouldnt. Havent rerun this one against the actual population/elevation rasters yet though (not something I can regenerate without redownloading the source data). So the population weighted percentages in the paper are still the old bbox only version for now. Flagged tht clearly in the Limitations section instead of changing numbers I havent actually recomputed.

### Where This Leaves Things

Central finding is completely unchanged. Seychelles still comes out as the highest compound risk island despite Maldives having higher raw physical exposure, nd tht's still robust across the full weighting sweep. Wht changed is Fiji's specific exposure nd compound score numbers (more accurate now, not less favorable to anything I was trying to protect), the governance correlation is now computed the right way instead of quietly including a placeholder zero, nd the coral significance test is running the statistically correct version for seasonal data. None of this came from rerunning any acquisition scripts against new data. It all came from actually rereading my own code against wht it claimed to do, same way the Winston/Yasa correction in Entry 8 did.

## Entry 10

Did another pass over the current `main` branch nd found 2 real internal consistency issues tht survived from before the Entry 9 cleanup. Both wording only. Neither one changes a reported number or a conclusion. But both were worth fixing since the whole point of this log is catching exactly this kind of drift.

**`dashboard/pages/7_Interactive_Maps.py`**, Lakshadweep's ecosystem overlay caption said mangroves *and* formal WDPA protection were "both confirmed absent, a finding independently confirmed across multiple data sources." Tht's correct for mangroves (confirmed absent, matches `mangrove_3point_comparison.py` only ever testing Maldives, Seychelles, and Fiji). But it's wrong for WDPA. `compute_wdpa_area.py` nd `wdpa_coastal_buffer.py` both say straight out tht Lakshadweep has no WDPA dataset available at all. Nd the paper itself (Section 3.7, nd again in Limitations) is careful to call this "a missing data gap, not a governance finding." This one dashboard caption was the only place in the whole project tht had drifted into claiming the missing WDPA data as a confirmed ecological absence instead. Reworded it to separate the 2 claims properly: mangroves confirmed absent, WDPA data simply unavailable.

**`research_paper_figures.py`**, the comment above the population weighted exposure figure called those numbers "population_weighted_exposure.py's validated final run." They arent. `population_weighted_exposure.py`'s own module docstring, nd the paper's Section 4.1 / Limitations text, both already say these are still the old bbox only v4 numbers, not yet rerun against the real rasters with the v5 boundary polygon mask. The paper text already said this. It was jst this one code comment tht was still describing the numbers as finished nd validated. Reworded the comment to match wht the docstring nd the paper actually say.

Neither fix touches a single reported number, a figure, or a conclusion. Both were jst internal consistency wording gaps, now closed.

## Entry 11

### Status

Finally got my hands on the real population nd elevation rasters for all 5 islands (they were sitting on my own laptop this whole time, I jst hadnt copied them into this checkout yet). So I reran population_weighted_exposure.py's v5 boundary polygon version properly for the 1st time. The numbers tht came out were wrong in a totally new way this time. Took a full afternoon of step by step checking to figure out why. Writing it all down here since this is exactly wht Entry 9 left as pending.

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## 1st Run Looked Very Wrong

So I ran the script nd got numbers way too low compared to real world population figures I already know. Maldives came out at only 6.5% captured. Lakshadweep at 7.3%. Real Maldives population is around 520,000 to 540,000 people, so tht 6.5% is nowhere close to right. Knew right away this was a new bug, not a real correction. Did NOT want to jst push these numbers into the paper without digging further.

## Ruled Out the Easy Explanations 1st

Checked the boring stuff 1st because tht's usually where the actual bug hides. CRS on the population raster nd the boundary files matched (EPSG:4326 both sides), so it wasnt a reprojection mixup. Then checked whether geometry_mask()'s default behaviour (only counts a pixel if its exact center point falls inside the polygon) was dropping population off narrow atoll islands thinner than one pixel. Added all_touched=True so it counts any pixel the polygon touches at all instead of jst the center. Helped a tiny bit (Maldives went from 6.5% to 6.8%, Lakshadweep from 7.3% to 6.3%, actually got slightly worse for Lakshadweep). But nowhere close to fixing the real problem.

Then checked if the boundary files were jst missing the big population centers entirely. Tested Male, Hulhumale, nd Kavaratti (the 3 biggest population centers across these islands) directly against the polygon geometries using a simple point in polygon check. All 3 came back True. Sitting right inside a polygon with basically zero distance to the boundary. So the boundary files do contain these specific islands. Tht theory was wrong too.

## Bug: Trying to Find Exactly Where the Population Was Disappearing

Ran a clean diagnostic split into 3 separate numbers. The raw total population in the whole raster (540,542 for Maldives, matches real world figures fine). The bbox only total with no polygon mask at all (also 540,542, exact same number, so the bbox window by itself isnt losing anything). Nd the bbox+polygon mask total (206,037). So the loss was happening 100% at the polygon mask step nd nowhere else in the pipeline. Only 16,516 out of 12,625,431 pixels in the whole window were even getting marked as land by the mask. Tht's 0.13%. Way too small for an island tht's supposed to have around 298 sq km of real land.

## Tested Whether the Boundary Polygons Were Jst Slightly Off Position

Next thought, maybe the polygons are basically right but shifted a little, so buffering them outward a small amount should recover most of the missing population fast. Tried buffering by 200m, 500m, 1000m, 2000m, nd 5000m nd watched how much population got captured at each step. If this was a small positioning bug I expected capture to jump up fast at a small buffer then flatten out. Instead it climbed slowly nd steadily all the way out to 5km (Maldives went 38% → 40% → 42% → 50% → 68% → 93%), never jumping anywhere. Tht kind of slow steady climb doesnt look like a simple offset bug. Looks more like the polygons are jst plain too small or missing pieces, nd buffering keeps bumping into more real (but unmapped) island area the further out u go.

## Checked If the Population Raster Itself Was Coarse nd Blurry

Had one more theory before blaming the boundary files completely. Wht if the population raster itself was originally coarser (like the common 1km WorldPop resolution) nd got resampled up to this 100m grid without proper handling. So the numbers technically sit on a 100m grid but really represent blurry ~1km blocks tht spill onto water. Zoomed into the actual pixel values right around Male nd Kavaratti to check this. Turned out

to be completely wrong. The values change smoothly nd realistically from pixel to pixel (110, 376, 498, 707, 689...), with zero pixels matching their neighbour 10 pixels away. This is real, fine grain data, not blurred blocks. So the population raster itself is fine nd this theory is dead too.

### Root Cause: The Boundary Polygon Files Are Just Incomplete

Finally jst measured the actual land area covered by the boundary polygon files nd compared it against real world island area figures. Tht's wht settled it. Maldives' boundary file only adds up to abt 118 sq km of total polygon area. Real Maldives land area is close to 298 sq km. So more than half of Maldives' real islands are jst not in this file at all.

Lakshadweep was a more confusing case at 1st. Its polygon file's total area (34.14 sq km) is actually close to the real number (abt 32 sq km), so I expected it to capture population fine. But it still only captured 13% of the real population even with all_touched=True on. Tht tells me the individual islands tht ARE in the file are drawn smaller or older than their real current built up extent (probably an old or generalised coastline, not matching wht's actually built nd populated today), rather than whole islands being missing the way Maldives is missing them.

Either way, both islands come back to the same root cause. The boundary shape files I have are jst not a reliable, complete map of where people actually live. The masking code is doing exactly wht it's supposed to do with wht it's given. It's the input data tht isnt good enough yet.

### Decision: Not Swapping v4 for v5 Numbers Yet

Pushing the current v5 output into the paper would replace a limitation I'd already disclosed (v4's bbox numbers, which slightly over count because a box can include some open water) with a WORSE nd undisclosed problem (v5 badly under counting real people, by around 60% for Maldives). Tht's a step backward, not a fix, even though the code itself is working correctly now.

So I'm keeping the v4 (bbox) numbers in the paper exactly as they are. Nothing in the actual reported table changed. Wht I did change is the wording in 3 places (population_weighted_exposure.py's own top note, the paper's Limitations section, nd the executive summary checklist), so all 3 now explain the real reason clearly. The boundary polygon files are incomplete, tht's been checked nd confirmed, nd fixing it properly needs a better set of island boundary shapes, not another round of code changes. This closes out wht Entry 9 left as an open item. Jst not with the ending I was hoping for. The plain answer this time is "found the real reason, still cant use it yet."

## Entry 12

### The Zero Reading Flag From Entry 9 Turned Out to Matter After All

Entry 9 flagged Maldives (99.0% of points reading exactly 0.0m) nd Lakshadweep (77.8%) as showing the same zero heavy pattern as Canary/Fiji, but left it as an open item since I hadnt done a pixel level check on either one. Finally went nd did tht check properly this time. Opened `data/terrain/maldives_elevation.tif` nd `lakshadweep_elevation.tif` in QGIS nd looked at the actual pixel grid, not jst the settlement sample points. Nd it's not jst 99% of settlement points reading zero, it's abt 99.99% of the ENTIRE raster reading exactly 0.0. Wall to wall. No gradient anywhere. Tht's not "low lying atoll terrain," tht's a coverage gap. The file never actually downloaded real elevation data for either island in the 1st place.

Redownloaded the underlying Copernicus DEM GLO-30 tiles directly from ESA's public archive (18 tiles for Maldives, 8 for Lakshadweep) nd rebuilt each island's raster from the complete tile set (`*_elevation_v2.tif`). 2 Maldives tiles nd 3 Lakshadweep tiles came back 404. A real gap in wht's publicly available, not something on my end. So even the corrected files still hve a residual coverage gap: 177 of 996 Maldives settlements (17.8%) nd 8 of 36 Lakshadweep settlements (22.2%) still sit in unretrievable tiles.

Since these 2 ARE real low lying coral atolls, I couldnt jst reuse the blanket "zero = bad reading" rule I used for Canary/Fiji. A real settlement here can legitimately sit at or near 0m, so dropping every exact zero would delete real at risk points right along with the artifacts. Instead built a 5x5 pixel neighborhood check. A 0.0 reading only gets treated as a void if its whole surrounding window is ALSO flat 0.0 (a signature no real terrain produces). Otherwise it stays as an actual reading. As a sanity check, looked up Malé's published average elevation (abt 2.4m, above the 1m threshold). Matches wht the corrected data now shows, not the original.

Recomputed against the remaining 819 Maldives nd 28 Lakshadweep settlements. Physical exposure moves from 99.1% nd 77.8% down to 14.5% nd 7.1%. This changes the paper's central result in a way Entry 9's Fiji fix nd Entry 11's population mask investigation never touched. **Seychelles, not Maldives, now has the highest settlement based physical exposure in the sample**, not jst the highest compound score. The "physical exposure alone would've picked the wrong island" framing I wrote back when I 1st got these numbers (see the entry above from way earlier) doesnt hold anymore. Exposure alone now points to the same island the compound score always did.

## Same Fix Applied to the Population Weighted Side, nd How Entry 11's Decision Got Undone by Accident

Since population_weighted_exposure.py samples the exact same elevation.tif files, it needed the identical fix. Switched Maldives/Lakshadweep over to `_elevation_v2.tif` nd added a raster level version of the same 5x5 void check (population sitting on a detected gap patch excluded from both numerator nd denominator, same as how Fiji's Lau Islands population is already handled).

1st run after this fix gave a number tht immediately looked wrong again. Lakshadweep's total population came out to only 8,893, against a real population near 64,000. Recognized the smell of this right away from Entry 11 nd went looking for wht I already knew about, the v5 boundary polygon incompleteness. Turned out Entry 11's decision to "keep v4, not v5" was only ever written down in comments nd the paper's wording. Never actually added code tht enforced it. `population_weighted_exposure()` was calling `load_boundary_geoms()` unconditionally whenever the boundary file loaded without throwing an outright exception, which it does every single time, despite the polygons themselves being known incomplete. So every run since Entry 11, including this one, was silently using the broken v5 masking instead of the v4 bbox method the paper actually reports. Entry 11 diagnosed the disease correctly. The prescription jst never made it into the actual script.

Fixed it properly this time with an explicit `APPLY_BOUNDARY_MASK = False` flag instead of a decision tht only lives in a comment. Reran: Seychelles, Fiji, nd Canary Islands (none of which this bug touched) came back identical to the numbers already in the paper (17.6%, 2.1%, 1.6%). Good sign the fix didnt disturb anything tht wasnt already broken. Maldives moved from an artifact inflated 64.5% to 16.9%. Lakshadweep from an artifact inflated 87.5% to 2.2%.

## Where This Actually Leaves the Central Finding

With both fixes in, all 3 independent measures (settlement based exposure, population weighted exposure, nd the compound vulnerability score) now agree Seychelles is the highest risk island. Tht's a real change from how I'd been framing this project. I used to argue "physical exposure alone would mislead u, the compound score catches wht exposure misses." Tht argument doesnt work anymore since exposure isnt misleading here either. The accurate version of the finding now is convergence, not correction. Every independent way of measuring risk lands on the same island. Went through nd reframed the paper, executive summary, README, nd every dashboard page tht referenced the old numbers to say this properly instead of leaving the disproven "exposure would mislead u" argument sitting next to numbers tht no longer support it.

Recomputed the downstream stuff tht depends on the compound score too. The governance correlation (H3) came out at r=0.965, p=0.035 (up from r=0.862, p=0.138), which crosses into conventional statistical significance for the 1st time. Still only n=4 though, so I'm not letting myself read "significant" as "settled." A single data point could still move both numbers a lot at this sample size, nd said so directly everywhere this result gets reported. The weighting sensitivity sweep also changed. The ~76.8% Maldives overtakes Seychelles crossover point from Entry 9 is gone entirely. Seychelles now leads across the full 0–100% range, no crossover anywhere, which is a cleaner robustness result than before.

Fixed every place carrying the old numbers tht I could find: `slr_exposure_analysis.py`, `population_weighted_exposure.py`, `compound_vulnerability_score.py`, `governance_correlation_test.py`, `research_paper_figures.py`, `precompute_elevations.py`, `export_settlement_elevations_full.py`, `build_interactive_plots.py`, the paper, the executive summary, the README, nd every dashboard page (`app.py` nd the numbered pages under `dashboard/pages/`) tht had the old Maldives/Seychelles ranking baked into its prose or chart data. None of this came from new data collection. Same as Entry 9, it all came from going back over my own code nd checking it actually did wht I assumed it did. This time starting from a QGIS pixel check I should've done back in Entry 9 instead of leaving it as an open item.

# Entry 13

## Checked Whether the Repo Was Actually in Sync

After the Entry 12 changes I wanted to make sure every place in the repo showed the new numbers. Wrote down wht could still be stale: old numbers in the paper, governance r=0.862 vs r=0.965, hardcoded numbers in governance_correlation_test.py nd build_interactive_plots.py, nd the old ~76.8% weighting crossover.

Cloned the repo fresh nd checked git log 1st, so I was looking at wht's actually on main. Then went thru each point by hand, running governance_correlation_test.py nd reading build_interactive_plots.py's code. Everything was already in sync. Commit d6280da ("Sync corrected findings") had fixed all of it. r=0.965 is the main number everywhere, r=0.862 only shows up once nd its labeled as the old pre correction estimate. The weighting sweep chart already says "no crossover" in its subtitle.

## Softened the Governance Significance Wording

Asked myself the harder question after tht. Not "is the repo synced" but "is this actually good enough for a real journal." The n=4 governance correlation is the weak spot a reviewer would flag fastest. r=0.965, p=0.035 is technically under 0.05, but calling it "reaches statistical significance" the way I had it worded sounds more confident than 4 data points can really back up. Went thru the paper, executive summary, nd README nd changed every instance of tht phrasing to "crosses the conventional 0.05 threshold" instead. Didnt touch a single number, jst the framing around it, so a reader doesnt walk away thinking this is settled when it isnt.

## Spatial Averaging the Coral DHW Series Instead of 1 Pixel

The coral thermal stress series was always sampled from a single representative coordinate per island, which I'd already disclosed as a limitation but never actually tested against a spatial average. Built a small box (±0.15°) around the same coordinate nd pulled the average across every valid pixel in tht box instead of jst 1 pixel, then reran the seasonal Mann-Kendall trend test on the new series.

Nd the answer held up. Maldives p=0.0011 (was 0.0046), Seychelles p=0.0194 (was 0.0069), both still significant. Fiji, Lakshadweep, nd Canary Islands all still come back not significant, same as before. Canary's actual mean value shifted a lot (0.714 down to 0.478, since the box around it only had 46% valid ocean pixels), but the trend conclusion didnt move at all. So this is a real, tested robustness check now instead of jst an acknowledged gap sitting in the limitations section.

## The Boundary Polygon Problem From Entry 11, Finally Actually Fixed

Entry 11 diagnosed this nd gave up on it. The boundary files were jst incomplete, nd fixing it properly needed better island shapes, not more code. Went back nd actually tried to get better shapes this time.

1st attempt only kept coastline ways tht were already closed loops on their own. Worked great for Maldives (304 sq km, real figure is abt 298), but came back WORSE than the old broken file for Seychelles, Fiji, Canary, nd Lakshadweep, all of them smaller than before. Turns out most real coastlines arent 1 closed way, theyre several segments joined end to end, nd only tiny simple atoll islets happen to be 1 closed way each.

2nd attempt tried also pulling in multipolygon relations, thinking bigger islands might be tagged tht way instead. Came back with 0 results every single time. Not tht kind of data here.

Realized the actual fix nd redid it properly. Fed every coastline way, open or closed, into shapely's polygonize() function, which stitches together anything sharing an endpoint into whatever closed loops it can find. Tht fixed Maldives nd Seychelles right away, both landed close to real figures. Fiji stayed broken tho, because I'd split it into 8 smaller download tiles to avoid server timeouts, nd Fiji's 2 biggest islands are bigger than any single tile, so their coastline ring got cut in half across 2 separate downloads nd never closed. Fixed tht by combining ways from every tile 1st, then running polygonize() once across the whole combined set instead of tile by tile. Canary nd Lakshadweep also needed a small padding buffer added around each search box so a ring jst barely touching the edge didnt get cut off. Tht padding accidentally pushed 1 of Fiji's tiles past 180 degrees longitude (invalid), which took another fix to clamp properly.

Even after all tht, Fiji stayed stuck at abt 13,000 sq km against a real figure near 18,274. Reran it with longer delays between requests thinking it was a rate limit issue, nd got the EXACT same number again, down to the decimal. Tht told me it wasnt a network problem, it was a real gap in the raw coastline data itself, small spots where 2 mapped segments jst dont technically touch. Switched to OSM's official pre built "land polygons" dataset for Fiji specifically, which already has this exact kind of gap cleaned up before its published. Tht got Fiji to 18,820 sq km, finally close to the real number.

Checked every island's final number against an actual cited source afterward instead of jst eyeballing it. Maldives 317.57 vs 300.00 (World Bank), Seychelles 449.55 vs 460.00 (World Bank), Fiji 18,820.57 vs 18,272 (Wikipedia geography page), Canary Islands 7,432.61 vs 7,447 (Wikipedia), Lakshadweep 30.84 vs 32 (the official Lakshadweep government site). All 5 within abt 6% of a real, citable number. Entry 11's "not fixed yet, needs better shapes" is finally closed out for real this time.

## Rerunning Population Weighted Exposure With the New Shapes

Swapped population_weighted_exposure.py over to point at the new boundary files nd flipped APPLY_BOUNDARY_MASK back to True (Entry 11 hd this forced off, for a very good reason back then). Running it now to see the actual new numbers. If it worked, Lakshadweep's total population shouldnt come out anywhere near the ~8,893 it did last time when the mask was silently broken, it should land somewhere close to the real ~64,000. Waiting on tht result before touching anything in the actual paper.

# Entry 14

## The Result Came Back... nd It Wasnt Wht I Expected

Ran the v2 population weighted script (the 1 pointed at the new, actually validated boundary shapes) nd waited. Lakshadweep 1st, since tht was the number I was most worried abt. It came back close to the real population, not tht broken ~8,893 anymore. So the mask itself is working now, tht part I was confident abt.

Then I looked at the full table properly nd my stomach kind of dropped a lil. Seychelles nd Maldives, on population weighted exposure specifically, had flipped. Maldives came out at abt 14.9%, Seychelles at abt 14.2%. Jst barely, but Maldives was ahead. Every single version of this paper, since the very start, has said Seychelles is the highest exposure island on every measure. Nd now 1 of those measures didnt agree anymore.

1st reaction was jst "did I break something." So I didnt jst update the paper. Had to choose between updating the paper straight away or verifying 1st. Picked verify 1st. This is exactly the kind of thing where jst trusting a new number bc it came out of a "fixed" script would be the wrong move.

## Checking It Myself Before Touching a Single Word of the Paper

Went back to raw basics. Pulled the actual population raster nd the new boundary file for jst Seychelles nd Maldives, wrote a tiny separate script tht redid the exact same masking logic by hand (geometry_mask, all_touched=True, same as the real script), nd summed up the population by hand.

Seychelles matched exactly. 94,175 people inside the mask, same number the real script reported, down to the digit. Maldives came out close too, abt 498,819, matching within a small margin. So the population side of this isnt some fluke or rounding bug, its reproducible from scratch.

Cldnt fully check the elevation/at risk half the same way tho, bc those raster files (nearly 1GB nd 1.5GB) were too big to pull across for this kind of quick check. So tht part wasnt checked by hand.

But the mechanism made sense too, not jst the numbers. The whole point of fixing the boundary mask was to stop counting open water next to an island as if it were populated land. Maldives is a spread out archipelago with a lot of water between its islands inside the old bounding box. So a tighter, more accurate boundary would naturally pull the Maldives number down less than expected, or in this case just enough for it to land ahead of Seychelles once Seychelles' own number also came down for the same reason. Not a bug. Jst the fix doing exactly wht it was supposed to do.

## Updating the Write-Ups

Once the verification above was done, I updated the write-ups.

So updated the paper, the executive summary, nd the README. Not jst swapping numbers in a table either. Had to go thru every sentence tht said "Seychelles is highest on every measure" or "all measures agree" nd

actually rewrite the claim to be accurate. Settlement based exposure nd the compound score still say Seychelles, tht part didnt change. Population weighted exposure specifically now says Maldives, narrowly, like a near tie (abt 0.7 percentage points apart) rather than some clean win. Wrote down the old number, the new number nd why, same as every other correction.

Table ended up like this in the end: Seychelles 14.2%, Maldives 14.9%, Lakshadweep 2.1%, Fiji 1.6%, Canary Islands 1.4%. All 5 numbers moved down from wht the paper hd before, which makes sense since the old bounding box version was always going to be counting some extra water as land.

## Dashboard Too

Realized halfway thru tht the Streamlit dashboard hd the exact same old numbers baked into it in a bunch of spots, the homepage, the physical exposure page, the compound vulnerability page. Even 1 of the actual chart images (fig3, settlement vs population weighted) was generated from a script tht still hd the old hardcoded 17.6/16.9/etc values in it. So fixed the dict in tht script too nd regenerated the chart image itself, not jst the text around it. Wouldve been a bit embarrassing to fix the paper nd leave the live dashboard showing the opposite ranking.

## Cleaned Up the Local Fix Scripts

Stripped the explanatory comments nd docstrings out of the 4 scripts I'd written for local runs (the coral area avg 1, the 2 boundary rebuild 1s, nd the population weighted v2ready 1). Those docstrings were mostly me explaining stuff to myself mid debugging, not really needed once the script jst works. Wrote a small script using Python's own tokenize nd ast modules to strip every # comment nd every real docstring cleanly without breaking the actual code, ran it, double checked each file still parses nd still does exactly wht it did before. Confirmed 0 leftover comments anywhere after.

## The Big Question Again: Is This Actually Ready

Went thru the whole project again, this time specifically looking for anything still out of sync now tht the population weighted numbers hd changed. Found 1 real problem this time, not jst a wording thing. The actual population_weighted_exposure.py sitting at the root of the repo, the real 1, not the local_fix_scripts copy, still hd APPLY_BOUNDARY_MASK set to False nd was still pointing at the old boundary files. So if anyone actually ran the canonical script in this repo right now, theyd get the old numbers, not wht the paper says. Exactly the kind of stale code vs paper mismatch I was checking for in Entry 13. Fixed it myself, flipped the flag, updated all 5 paths to the new boundary files, nd rewrote the docstring to explain the full history instead of jst deleting it.

Also did some actual research on the journal itself this time, not jst guessing. Ocean and Coastal Research turns out to not be restricted to Brazil or South America like I half assumed, its actually international in scope, Scopus nd DOAJ indexed, single blind peer review, no charge to publish. No official Web of Science listing tho, so its a solid mid tier journal, not a flagship 1.

Found 2 more real gaps while scanning. 1st, the coral DHW spatial averaging robustness check I'd already done earlier was never written into the paper. The paper still said "this is future work" in 3 different places, which was jst flat out wrong, I hd already done it nd it hd already held up. 2nd, the land area numbers I used to validate the boundary shapes, World Bank nd Wikipedia nd the Lakshadweep gov site, were named in the text but never actually added as real references in the References list.

### Closing Both of Those Out Right Now

For the coral 1, went nd reran the actual numbers myself 1 more time to be sure before writing anything down, since I didnt want to just copy from memory nd get a digit wrong. Confirmed Maldives p=0.0011, Seychelles p=0.0194, both still significant under the spatial average, nd Fiji/Lakshadweep/Canary all still not significant (p=0.517, 0.810, nd 0.382). Every 1 of them matched the original single point conclusion exactly. Rewrote the 3 spots in the paper nd 2 spots in the executive summary tht said "future work" to instead say wht actually happened nd wht the result was.

For the references, swapped out the informal "Wikipedia" mentions for real sources. Fiji nd Canary Islands land area now cite the World Bank nd Instituto Canario de Estadística instead, since Wikipedia isnt really something u cite in an actual paper. Added all 5 as proper entries in the References section, nd updated the numbers in the text itself to match the real, slightly more precise figures from those sources (298 for Maldives instead of 300, 18,270 for Fiji instead of 18,272, tht kind of small correction).

Still left to do after all this: the actual journal formatting pass, which is still deliberately sitting at the very end of the list where it belongs, same as always.

## Entry 15

### Status

Went back through the whole project (10 October 2026), followed by code and documentation fixes that did not require rerunning the analysis. The analysis itself still has to be rerun with the corrected code (see "Still to run" below); until then the exposure, compound-score and governance numbers in the paper, README, Executive Summary and dashboard should be treated as superseded.

### What I found

- **Seychelles exposure was a NoData artifact.** 188 of the 244 Seychelles settlements read −32767 m, the elevation raster's NoData value, and were counted as "at or below 1 m". The Seychelles raster is 99.97% NoData (its own statistics file reports 0.034% valid pixels). On the 56 settlements with a real reading, exposure is 3 of 56 (5.4%). This is also why Seychelles showed the same 78.3% at the 0.5, 1.0 and 1.5 m thresholds.
- **Lakshadweep's 2 at-risk settlements were outside the DEM.** North Islet and Sand Cay (about 12.35°N) lie north of the re-downloaded DEM, read 0.0, and passed the 5×5 void test. From the available tiles Lakshadweep exposure is 0 of 26.
- **The "404 tile" explanation was wrong.** Rebuilding the Maldives DEM from the raw tiles reproduced the 177 excluded / 819 kept / 14.5% result exactly, but none of the 177 excluded points lie in a missing tile. Copernicus only publishes tiles that contain land; the excluded points are settlement nodes on 0 m water pixels. 96 of the 119 Maldives at-risk points are also exact 0.0 water pixels.
- **Downstream effects.** With Seychelles corrected, the compound score of Seychelles (0.68) and Maldives (0.65) are nearly tied and the top island changes at about 52% exposure weight; the governance correlation falls from r = 0.965 (p = 0.035) to r = 0.75 (p = 0.25). Even as reported, removing Seychelles alone gives r = 0.51.
- **NDBI encroachment.** The bounding boxes are 99.7% (Maldives) and 85% (Seychelles) ocean, and the Statistical API values had no cloud or water mask. The project's own cloud-masked NDBI rasters for the same boxes give a land-only change of −0.0003 for Seychelles, not +0.1135.

- **WDPA coverage.** The coastal ratio used the old, incomplete boundary files and summed overlapping WDPA polygons (overlaps inflate Seychelles by about 57% and Canary by about 82%).
- **Fiji settlement coordinates.** Six Fiji settlements (Lau Group, Vanua Levu Group, Taveuni, Vanua Levu, Korolevu Island, Rabi) were stored at about 0° longitude because their point was averaged across the 180° meridian.
- **Verified as correct:** all seasonal Mann-Kendall p-values (single cell and box average), the DHW period changes, and the mangrove areas reproduced exactly.

## Code changes

- New `scripts/common/dem_sampling.py`: one DEM rule for all five islands. Each settlement is classified as valid, snapped_to_land (point on a 0 m water pixel; elevation = median of land pixels within about 60 m), water_no_land_nearby, nodata or out_of_extent. NoData and out-of-extent points can no longer be counted as at risk. It also corrects the six Fiji longitudes.
- `slr_exposure_analysis.py` rewritten on that module; reports coverage and status counts per island, the main rule and a strict rule (water-pixel points excluded), and writes per-settlement CSVs. `export_settlement_elevations_full.py` and `scripts/processing/precompute_elevations.py` now call the single implementation; `dashboard/precompute_elevations.py` uses the same module.
- `population_weighted_exposure.py`: NoData, NaN and 0 m water pixels are removed before resampling (average resampling of land pixels only); population with no usable elevation is excluded and reported; large windows are processed in 1° chunks; results go to `data/population_weighted_exposure.csv`.
- New `scripts/acquisition/download_copernicus_dem_tiles.py`: downloads GLO-30 tiles for any island over the full settlement extent and builds a VRT mosaic (Seychelles is too large for an in-memory merge).
- `wdpa_coastal_buffer.py`: validated v2 boundaries, dissolved WDPA polygons, a new share-of-coastal-zone metric, a strict IUCN I–IV variant, and Lakshadweep reported as missing rather than zero. `filter_wdpa_islands.py` now also reads the WDPA points file.
- `compound_vulnerability_score.py` and `governance_correlation_test.py` read their inputs from CSVs (no hand-typed numbers), report the weighting crossover, Spearman's rho and leave-one-out correlations.
- `coral_trend_test.py`: adds calendar-month seasonal Mann-Kendall, annual-maximum Mann-Kendall and a results CSV. New `download_dhw_reef_cells.py` averages DHW over the 5 km cells that contain mapped reef.
- New `ndbi_land_change.py`: land-only NDBI change from the cloud-masked rasters, near-ecosystem zone, block-bootstrap interval.
- Figures: coral figure redrawn as one panel per island with annual maxima; governance figure no longer plots Lakshadweep at zero; weighting-sweep subtitle computed from the data; population figure reads the CSV; new `study_area_map.py` draws the study areas from the project's own data. SLR webmaps show settlements without a usable elevation in grey with the reason, instead of "−32767 m".
- New `scripts/run_all.py` runs the full pipeline in order.

## Documentation changes

Research question and hypotheses stated in Section 1 of the paper; study-area map added as Figure 1; corrected the 404 explanation, the GMW epoch description, the seasonal-test description and the "24-year" wording (the series covers 25 calendar years, 1996–2020); Canary Islands disclosed as having no mapped reefs;

conclusion no longer asserts misallocation; image paths fixed for the `paper/` folder. Executive Summary no longer claims that exposure alone misranks islands. CITATION.cff and dashboard titles no longer say "island nations". New `DATA_SOURCES_AND_LICENSES.md`.

### Still to run (needs the full rasters and Python environment)

1. `python scripts/acquisition/download_copernicus_dem_tiles.py --all`
2. `python scripts/acquisition/download_dhw_reef_cells.py`
3. `python scripts/processing/filter_wdpa_islands.py` (WDPA points check for Lakshadweep)
4. `python scripts/run_all.py`
5. Rewrite the numbers in the paper, Executive Summary, README, CITATION.cff and dashboard from the new CSVs.

## Entry 16

### Status

Reran everything with the Entry 15 fixes plus a bare earth elevation model (10–11 October 2026). Numbers are now frozen. The paper, Executive Summary, README, CITATION.cff, dashboard and both portfolio pages were rewritten from the new CSVs, nd the dashboard now reads its numbers straight from `data/` so it can't go stale again.

### Why a Second Elevation Model

Entry 15 fixed the NoData problem, but the 1st clean run on Copernicus GLO-30 still bothered me. Copernicus is a surface model. On a palm covered atoll it measures the top of the canopy nd the roofs, not the ground, nd my whole threshold is 1 m. So I added DeltaDTM v1.1 (Pronk et al. 2024), which is Copernicus with trees nd buildings removed using ICESat-2 nd GEDI lidar. Same 1 arc second grid as Copernicus, heights above EGM2008, about 0.4–0.5 m error against airborne lidar.

The 4TU download page was down for storage maintenance, so I exported the tiles from the Earth Engine community catalogue copy instead (`projects/sat-io/open-datasets/DELTARES/deltadtm_v1-1`): 77 tiles, one Drive export each, with `scripts/acquisition/export_deltadtm_gee.py`. Tasks didn't show up in the Code Editor because I'd started them from VS Code, so I downloaded from Drive by hand nd checked the folder until all 77 were there. `prepare_deltadtm_tiles.py` checks each tile's bounds against the Copernicus tile it replaces nd builds one VRT per island. It's now the 1st step of `run_all.py`.

### The Seychelles Granite Hill Bug

1st DeltaDTM run, a few Seychelles points looked wrong when I opened the per point file. Petite Sœur, Grande Sœur, Île Marianne: Copernicus 30–76 m, DeltaDTM close to 0 m. These are small forested granite islands. A 50 m tree canopy is not a thing there, so DeltaDTM was reading a hill as sea level. DeltaDTM is clipped at 30 m nd only meant for the coastal zone, so I added one rule: where Copernicus is above 30 m, DeltaDTM is not used nd the point counts as high terrain. Same rule in the population script. Also: where DeltaDTM is empty nd Copernicus is above 10 m, tht's "above 10 m", which is wht an empty cell means in this product. Reran once more, then froze the numbers.

### Coral: Mean Over Reef nd the Serial Correlation Correction

The coral side changed too. Instead of 1 5 km cell per island, the series is now the mean over every DHW cell tht touches mapped reef (`download_dhw_reef_cells.py`). And the seasonal Mann-Kendall test is now the version corrected for serial correlation (Hirsch & Slack 1984), on calendar month means, 300 months. DHW is a 12 week running sum, so one warm spell shows up in 3 nd 4 months in a row, nd the plain test counts each of them as new evidence. With the correction, no island is significant: Seychelles p = 0.14 (uncorrected 0.0008), Maldives 0.31, Fiji 0.55, Lakshadweep 0.90. "Significant for Maldives nd Seychelles" had been a headline result since version 1. But the uncorrected p is wrong for this kind of series. Had to fix a small bug on the way: the monthly series came out all NaN because the reef cell times were timezone aware nd the index wasn't.

## Decisions (frozen)

1. People (WorldPop) at or below 1 m is the main exposure measure. OSM points are mostly island nd islet names, so they're kept as a 2nd measure, renamed "named places".
2. DeltaDTM is the main elevation model; Copernicus is reported next to it everywhere.
3. Coral = mean DHW over reef cells + corrected seasonal Mann-Kendall.
4. Canary Islands' reef input to the compound score is 0 (no reef).
5. Compound score is descriptive. The top island depends on the weighting nd I say so.
6. Governance is descriptive (n = 4).
7. NDBI is supplementary, land pixels only.
8. Cyclone tracks (IBTrACS) dropped from the analysis; the Sentinel-2 record doesn't reach the big cyclones anyway.
9. EGM2008 is treated as mean sea level nd this is stated as an approximation.

## Final Numbers

People at or below 1 m, DeltaDTM (Copernicus in brackets): Maldives 11.82% (8.88), Lakshadweep 6.40% (0.79), Seychelles 2.07% (0.43), Fiji 0.28% (0.21), Canary Islands 0.19% (0.15). Named places at or below 1 m on DeltaDTM: Maldives 28.8%, Seychelles 24.7%, Fiji 4.7%, Lakshadweep 3.8%, Canary Islands 0.26%.

Reef heat stress change, 1996–2000 to 2016–2020: Seychelles +0.52, Fiji +0.29, Maldives +0.07, Lakshadweep +0.05 °C-weeks. Mangroves flat (Fiji 485.7 → 488.4 $km^2$).

Compound score at equal weights: Seychelles 0.581, Maldives 0.512, Lakshadweep 0.267, Fiji 0.256, Canary 0.000. The top flips to Maldives at 54% exposure weight. So the "Seychelles is top at every weighting" result from version 1 is gone, It was only ever true because both of version 1's inputs happened to be highest for Seychelles.

Governance: share of the 10 km coastal zone protected, Seychelles 71.4%, Canary 50.4%, Fiji 12.2%, Maldives 1.3%, Lakshadweep no record. r = −0.01 (95% interval −0.96 to 0.96), without Seychelles −0.95. The 0.965 from version 1 is gone.

NDBI on land: Maldives +0.055 (0.040 to 0.072), Seychelles −0.001, Fiji −0.011.

## What I Changed in the Write-Ups

- Dashboard: new `dashboard/results.py` loads every number from the CSVs; all pages rewritten on top of it. Trends page now shows the reef cell mean with period means instead of OLS lines. The slider data (`dashboard/static/settlement_elevations.csv`) is rebuilt by `run_all.py`.

- Paper rewritten for version 2, with a new Section 2.5 on surface vs bare earth models nd a Section 6.1 tht lists every version 1 result tht didn't hold nd why.
- Executive Summary, README, CITATION.cff (version 2.0.0) nd `DATA_SOURCES_AND_LICENSES.md` (DeltaDTM added) rewritten.
- Figure footers said "DeltaDTM v2"; corrected to v1.1.

### Still To Do

Re-export the three PDFs from the new markdown, push, nd tag v2.0.

## Entry 17

### Status

Second full check of the whole project (20 October 2026), followed by corrections that did not need the large rasters. The exposure, coral and mangrove numbers did not change. The compound score, the weighting crossover and the governance correlation did.

### Corrections

| Change | Before | After | Why |
|---|---|---|---|
| Compound score scaling | Coral input scaled across the 4 reef islands, Canary set to 0 afterwards | Both inputs scaled across all 5 island groups (Canary raw coral input 0) | The paper (Section 3.6) said five; the code did four, so Lakshadweep got the same coral input (0) as an island with no reef |
| Compound scores (equal weights) | Seychelles 0.581, Maldives 0.512, Lakshadweep 0.267, Fiji 0.256 | Seychelles 0.581, Maldives 0.563, Lakshadweep 0.320, Fiji 0.282 | Follows from the scaling fix |
| Top-island crossover | 54% exposure weight | 52% | Same |
| Governance correlation | r = −0.01 (−0.96 to 0.96), LOO up to 0.55; ratio r = 0.64 | r = −0.10 (−0.97 to 0.95), LOO up to 0.42; ratio r = 0.57 | Uses the corrected scores |
| "Uncorrected" coral p-values | Ordinary seasonal test on the raw | Ordinary seasonal test on the same | Like-for-like comparison; only the serial-correlation correction now differs |

| Change | Before | After | Why |
|---|---|---|---|
| | 30-day samples (Seychelles 0.0008, Maldives 0.07) | 300 monthly means as the corrected test (Seychelles 0.002, Maldives 0.049) | |
| Land-area validation (Section 6.3) | Maldives 317.6 km², Seychelles 449.6, Fiji 18,820.6, Canary 7,432.6 | 292.9, 443.9, 18,819.3, 7,434.1 (EPSG:6933, overlaps merged) | The earlier figures were not reproducible from the current boundary files; new `scripts/validation/boundary_area_validation.py` writes them. All still within 4% of the reference figures |
| DeltaDTM description | "cells above 10 m are left empty" | Coastal product meant for land below about 10 m, values clipped at 30 m | The tiles hold values up to 30 m |
| Named places, two DEMs | Not stated that DeltaDTM drops some points | Stated; on the common point set Seychelles still leads on Copernicus (8.4% vs 4.4%) | Different denominators (Seychelles 154 vs 162) |
| NDBI | "for the 3 islands" | The three windows named (Malé area, Mahé, western Viti Levu); possible effect of Hulhumalé reclamation noted | The analysis never covered whole countries |

| Change | Before | After | Why |
|---|---|---|---|
| WDPA note | – | 25 of the Maldives' 93 records are UNESCO-MAB zones, stated | Coverage definitions differ |
| Figures | Clipped subtitles (mangrove, compound), "SLR threshold" axis label, red/blue classes on the compound bars, OLS line through 4 points | Fixed; one colour; line removed | Presentation |
| Dashboard | v1 research question and island descriptions, "independent time points", outdated code screenshots, dropped cyclone material, figure numbers different from the paper | Brought in line with the paper; code screenshots hidden | Consistency |

### Still To Do

Rerun the light scripts, re-export the three PDFs, push, and redeploy the dashboard.

## Entry 18

## Status

New analysis (23 October 2026): do the statutory coastal setbacks of the five governments cover the buildings that stand on low land? Added as Sections 3.9 and 4.8 of the paper, with Figure 9. Nothing else in the results changed.

## Why

The governance part of the study only asked whether protected area coverage follows risk, with n = 4. A setback is the one coastal rule that decides where people may build, and buildings give hundreds of thousands of observations instead of four. The 100 m WorldPop grid is too coarse for a 20–25 m rule, so I used building footprints.

## What I Did

- Downloaded Overture Maps buildings (release 2026-08-19.0) for the five study areas (`scripts/acquisition/download_buildings_overture.py`) and reduced them to one representative point per footprint (`scripts/processing/export_building_points.py`): Maldives 124,936, Lakshadweep 9,464, Seychelles 53,565, Fiji 284,279 (OpenStreetMap only), Canary Islands 880,740.
- `scripts/analysis/setback_calibration.py` samples elevation with the same DeltaDTM/Copernicus rule as the population analysis, measures distance to the edge of the validated island polygons, and compares it with each setback. Added to `run_all.py`; it skips if the building files are absent.
- Figures: `outputs/plots/fig9_setback_calibration.png` (coverage curve, written by the analysis script) and `fig10_setback_close_ups.png` (`scripts/figures/setback_example_maps.py`, close-ups of the densest low-lying cluster per island); interactive `setback_coverage_curve.html` added to `build_interactive_plots.py`, whose compound bars are now one colour.
- Dashboard: new "Coastal Setbacks" tab on the Governance & Encroachment page, a policy-extension card on the home page, the setback question on Study Design, and buildings and legal sources on Methodology & Data. README, Executive Summary (version 2.1), CITATION.cff (2.1.0), DATA_SOURCES_AND_LICENSES.md (Overture, ODbL) and requirements.txt (pillow, overturemaps) updated.
- Checked the legal texts: Ley 22/1988 de Costas art. 23 (100 m, 20 m on urban land); Seychelles S.I. 91 of 2023 reg. 13 (25 m from the high water mark, 10 m with permission); Maldives tourism environment regulation cl. 2.10 (5 m from the vegetation line, resort islands only; nothing found for inhabited islands); Lakshadweep ICRZ 2019 20 m no development zone (secondary sources only, primary text still to check); Fiji none found.

## Results

| Island group | At or below 1 m | Setback | Low buildings outside every setback | 90% of low buildings within |
|---|---|---|---|---|
| Maldives | 13.4% | none (inhabited) | 99.4% | 225 m |
| Lakshadweep | 4.7% | 20 m | 93.2% | 345 m |

| Island group | At or below 1 m | Setback | Low buildings outside every setback | 90% of low buildings within |
|---|---|---|---|---|
| Seychelles | 4.6% | 25 m | 92.0% | 427 m |
| Fiji | 0.5% | none | 99.8% | 463 m |
| Canary Islands | 0.02% | 100 m / 20 m | 15.1% | 122 m |

The building shares follow the population shares from Section 4.1, which is a useful cross-check. 45% of the low Seychelles buildings and 39% of the Fijian ones are on DeltaDTM cells below 0 m (south-east Mahé, Praslin, La Digue, Kadavu); dropping them changes the "outside every setback" share by under three points.

### Still To Do

- Check the primary text of the ICRZ 2019 notification for the Lakshadweep width.
- Look at the 4,478 Maldivian buildings that fall more than 50 m outside the island polygons (largest cluster in Addu Atoll); some islands may be missing from boundaries_v2.
- Carry the new section into the Executive Summary, README, dashboard and portfolios.

## Entry 19

### Status

Validation checks (23 October 2026). Went through the parts of the analysis that rest on my own code and checked them against independent calculations. Nothing in the results changed.

### What I Checked

- **Coral trend p-values.** Wrote the seasonal Mann-Kendall test and the Hirsch-Slack correction again from scratch in NumPy (`scripts/validation/coral_trend_check.py`) nd ran it on the same monthly series. The uncorrected p-values match pyMannKendall exactly. The corrected ones match to within 0.005 (Maldives 0.316 vs 0.313, Seychelles 0.144 vs 0.143, Fiji 0.553 vs 0.550, Lakshadweep 0.905 vs 0.900). None of them comes near 0.05, so H1 stays where it was.
- **Setback distances.** The setback script measures distance on a flat approximation around each island's mean latitude. Checked about 13,000 sampled buildings against ellipsoidal distances (Vincenty) to the same shoreline (`scripts/validation/setback_distance_check.py`). Median difference: 0.07 m Maldives, 0.17 m Lakshadweep, 0.04 m Seychelles, 0.40 m Fiji, 3.8 m Canary Islands. Fewer than 0.1% of sampled buildings change side of a 20, 25 or 100 m line.
- **Boundary reader.** The setback script reads the GeoPackages with its own small parser instead of GeoPandas. The land areas it gives (292.9, 443.9, 18,819.3, 7,434.1, 30.8 km$^2$) are the same as the ones `boundary_area_validation.py` gets with GeoPandas, so the parser reads the polygons correctly.
- **Elevation rules.** The named places, population nd building analyses read elevation in different ways (point, 100 m cell, building point), but they use the same rules for NoData, 0 m water cells, the 10 m nd 30 m Copernicus cut-offs nd negative values. Wrote them down as one table in Section 3.3 of the paper.

### Other Changes

- `run_all.py` now ends with "COMPLETE RUN" or "PARTIAL RUN" nd exits with an error code if any step failed, so old outputs can't be mistaken for fresh ones. The two validation scripts are part of the run.
- Paper: one sentence on what the study adds (Section 1), the elevation rule table (3.3), the coral check (3.5), nd the distance check, the mapped-boundary wording nd the one-point-per-building note (6.7).